A Survey of On-Policy Distillation for Large Language Models

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Abstract

Knowledge distillation has become a primary mechanism for transferring capabilities from frontier Large Language Models (LLMs) to smaller, deployable students. The dominant paradigm remains *off-policy*: students train on static, teacher-generated data and never encounter their own errors during learning. This train–test mismatch, an instance of *exposure bias* rooted in imitation learning theory, causes prediction errors to compound autoregressively at inference time with expected discrepancy scaling as $O(\epsilon T^2)$. On-Policy Distillation (OPD) addresses this by letting the student generate its own trajectories and receive teacher feedback on these self-generated outputs, reducing compounding to $O(\epsilon T)$. Despite rapid growth spanning divergence minimization, reward-guided reinforcement learning, and self-play, the OPD literature remains fragmented across communities with no unified treatment. This survey provides the first comprehensive overview of OPD for LLMs, covering over 100 methods from 2023–2026. Our contributions are fourfold: (1) We formulate a *unified f-divergence framework* over student-sampled trajectories, revealing how GKD, MiniLLM, DistiLLM, and G-OPD instantiate the same underlying objective under different parameterizations. (2) We organize the landscape around a *practitioner's decision chain* rather than descriptive categories, classifying each method by its core functional contribution: objective design, signal source architecture, or training dynamics optimization. (3) We provide a systematic *theoretical analysis* of success conditions, failure modes (flawed prefix traps, self-play saturation, calibration-capability gaps), and the formal equivalence between OPD and KL-constrained reinforcement learning. (4) We chart *open problems* including distillation scaling laws, uncertainty-aware feedback, agent-level distillation, and the dissolving boundary between knowledge distillation and RL.

Awesome List & Paper Collection
https://github.com/nick7nlp/Awesome-LLM-On-Policy-Distillation

1 Introduction

Large Language Models (LLMs) have reshaped natural language processing and, increasingly, artificial intelligence as a whole. Scaling from hundreds of millions to hundreds of billions of parameters, frontier models now demonstrate strong capabilities in reasoning, code generation, multilingual understanding, and instruction following (Yang et al., 2025a; DeepSeek-AI et al., 2025; Gemma Team et al., 2024). Yet the sheer computational cost of training and serving these frontier models remains prohibitive for most deployment scenarios, making the transfer of capabilities from large teachers to compact students not merely desirable but economically necessary. Knowledge distillation (KD), first formalized by Hinton et al. (2015) as training a student network to match the softened output distribution of a teacher, has since grown from a niche compression technique into a central pillar of the modern LLM development pipeline. The release of DeepSeek-R1 (DeepSeek-AI et al., 2025), which successfully transferred complex chain-of-thought reasoning from a 671B mixture-of-experts teacher into dense student models ranging from 1.5B to 70B parameters, illustrates how distillation now serves as a general-purpose *capability transfer engine* rather than a mere size-reduction tool.

As the role of distillation has expanded, so has awareness of a fundamental bottleneck in its dominant form. Conventional LLM distillation is *off-policy*: the student model p_θ learns to replicate the teacher's next-token distribution over a static corpus, typically sampled from the data distribution p_{data} or generated in advance by the teacher. During training, the student is teacher-forced, conditioning its predictions on flawless prefixes. During inference, it must generate autoregressively, conditioning each new token on its own (and potentially erroneous) outputs. This discrepancy between the training distribution and the student's own generation distribution creates a *train-test mismatch* that compounds over the length of the sequence. Formally, this phenomenon is an instance of *exposure bias* (Ross et al., 2011): a policy trained only on expert-state demonstrations drifts into unfamiliar states at test time and lacks the supervisory signal to recover. For autoregressive LLMs, the problem is particularly severe because errors at early positions propagate through the entire remaining sequence with expected discrepancy scaling as $O(\epsilon T^2)$ over a trajectory of length T, and the student receives no gradient information about how to behave in these self-induced off-distribution states. Gudibande et al. (2023) provided an influential early warning: imitating proprietary LLMs through off-policy distillation on API-generated data produces models that mimic surface style while failing to transfer genuine reasoning capability, with performance gains largely confined to tasks well-represented in the imitation data.

On-Policy Distillation (OPD) offers a principled solution. Rather than training on a static dataset, OPD shifts the sampling distribution from the fixed corpus to the student's evolving policy. The student generates its own trajectories, explores sub-optimal states, and queries the teacher for corrective feedback on those specific states. This feedback can take diverse forms: full token-level probability distributions in white-box settings (Agarwal et al., 2024; Gu et al., 2024), scalar rewards or pairwise preferences when only black-box teacher access is available (Ye et al., 2025; Jiang et al., 2023), and self-generated contrastive signals in teacher-free frameworks (Chen et al., 2024). The theoretical motivation draws directly from interactive imitation learning: the DAgger algorithm (Ross et al., 2011) established that querying an expert on the learner's own visited states reduces compounding error from $O(T^2)$ to $O(T)$. OPD instantiates this principle for autoregressive language generation, transforming distillation from one-shot distribution matching into an iterative, self-correcting optimization loop.

This approach has catalyzed rapid methodological innovation. Generalized Knowledge Distillation (GKD) (Agarwal et al., 2024) introduced on-policy sampling with configurable mixture ratios between student and teacher sequences. MiniLLM (Gu et al., 2024) recast the objective by minimizing Reverse KL divergence to avoid the mode-covering pathology of Forward KL. DistiLLM (Ko et al., 2024) proposed skewed KL objectives that stabilize optimization by mixing student and teacher distributions in the divergence argument. G-OPD (Yang et al., 2026c) proved the formal equivalence between on-policy distillation and dense KL-constrained reinforcement learning. On the self-distillation side, SPIN (Chen et al., 2024) demonstrated that a model can improve by distinguishing its own generations from human references without any teacher, while OPSD (Zhao et al., 2026b) showed that a model can distill from a privileged-information-conditioned version of itself. The field has grown explosively: while fewer than 10 papers addressed on-policy LLM distillation before 2024, over 90 methods have appeared in the subsequent 18 months, driven by both practical demands of efficient deployment and the theoretical realization that the boundary between distillation and reinforcement learning is dissolving. Industrial adoption has been equally rapid: Qwen3 (Yang et al., 2025a), DeepSeek-R1 (DeepSeek-AI et al., 2025), Gemma 2 (Gemma Team et al., 2024), and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) all incorporate OPD as a core training ingredient.

Despite this rapid growth, the OPD literature remains fragmented. Methods originating from the knowledge distillation community, the RLHF community, and the imitation learning community often address the same underlying problem with different formalisms, evaluation protocols, and terminology. Existing surveys of LLM distillation (Xu et al., 2024) predominantly organize the field around the classical compression framing, treating off-policy and on-policy approaches as interchangeable variants rather than as fundamentally distinct paradigms with different theoretical guarantees and failure modes. To date, no survey has provided a unified mathematical treatment of the on-policy dynamics that connect these seemingly disparate lines of work, nor has any systematically compared the trade-offs among white-box, black-box, and teacher-free instantiations of on-policy learning for LLMs.

This survey fills that gap. We provide the first dedicated and comprehensive treatment of on-policy distillation for Large Language Models, organized around a practitioner's decision chain rather than descriptive categories. Our contributions are as follows:

- **A Unified Theoretical Framework.** We formulate the transition from off-policy to on-policy distillation through the lens of sequential decision-making and f-divergence minimization over student-sampled trajectories (Section 2). This framework reveals how GKD (Agarwal et al., 2024), MiniLLM (Gu et al., 2024), and DistiLLM (Ko et al., 2024) instantiate the same underlying objective under different divergence choices, argument orderings, and sampling mixture coefficients, providing a common analytical language for methods previously studied in isolation.
- **A Decision-Centric Taxonomy.** We organize OPD methods by their core functional contribution along a practitioner's decision pipeline (Section 3): objective function design (Section 4), signal source architecture (Section 5), and training dynamics optimization (Section 6). This strict one-method-one-category classification directly addresses the "which method should I use?" question that practitioners face when selecting among the growing number of available methods.
- **Theoretical Analysis of Failure Modes.** We synthesize empirical findings into a coherent understanding of when and why OPD breaks down (Section 7), including the flawed prefix trap, self-play saturation, diversity collapse, and the calibration-capability gap, and provide formal conditions under which OPD provably improves over off-policy alternatives.
- **Comprehensive Coverage and Actionable Guidance.** We survey over 100 methods spanning white-box, black-box, and self-distillation regimes, with three detailed comparison tables (Tables 1, 2, and 3) and a concrete decision tree (Section 3.3) mapping deployment constraints to recommended methods and compute budget guidelines.

The remainder of this survey is organized as follows. Section 2 establishes mathematical preliminaries, formalizing autoregressive generation as a sequential decision problem and deriving the exposure bias that motivates on-policy approaches. Section 3 presents our practitioner-centric landscape and decision tree. Sections 4–6 provide an in-depth analysis of OPD methods organized by the three stages of the decision pipeline: objective function design, signal source architecture, and training dynamics optimization. The ordering of these sections is not arbitrary: each addresses a design decision that logically precedes the next, tracing the arc from "what to optimize" through "where the signal comes from" to "how to make training work in practice." Section 7 consolidates theoretical perspectives on success conditions, failure modes, and the formal equivalence between OPD and constrained RL, showing that the empirical patterns across Sections 4–6 follow from a small number of unifying principles. Section 8 discusses industrial deployments, system-level optimization, and emerging domains. Section 9 identifies open problems and future directions.

This survey covers over 100 distinct methods from 2023–2026, drawn from top venues (ICLR, ICML, NeurIPS, EMNLP, ACL) and recent preprints, with approximately 60% appearing after January 2025. We focus on methods where the student generates its own training data during the distillation process, including methods from adjacent fields (imitation learning, online RL, preference optimization) when they operate on student-generated data with teacher or verifier supervision. We exclude general off-policy KD, model compression techniques orthogonal to the training paradigm (pruning, quantization), and inference-time methods that do not modify model weights.

Notation. Throughout, p_T denotes the teacher distribution, p_θ the student parameterized by θ, π_{mix} the sampling policy for on-policy generation (which may differ from p_θ for mixed-policy methods), and $D_{\text{KL}}(P\|Q)$ the Kullback-Leibler divergence with the convention that P is the reference and Q is the approximation. When we write $y \sim p_\theta(\cdot|x)$, we mean the full autoregressive rollout generated token-by-token from the student's current policy.

2 Background and Unified Math

The necessity of OPD stems from the compounding geometry of autoregressive generation. This section formalizes the distribution mismatch, establishes the unified f-divergence framework that subsumes modern OPD objectives, and traces the historical arc from classical KD to modern on-policy methods.

Defining "on-policy." A distillation method is *on-policy* if the training data for the student is sampled from the student's own current policy p_θ at training time, rather than from a fixed external corpus $\mathcal{D}$ or from the teacher's generation distribution p_T. Formally, on-policy training optimizes:

$$\min_\theta \mathbb{E}_{x\sim\mathcal{D}}\, \mathbb{E}_{y\sim p_\theta(\cdot|x)}\, [\mathcal{L}(y, x; \theta, T)] \tag{1}$$

where $\mathcal{L}$ is the distillation loss (divergence, reward, or hybrid). The critical distinction is that the *outer* expectation is over the student's own generations, not a static dataset. This creates a non-stationary optimization landscape: as θ updates, the data distribution p_θ shifts, requiring fresh rollouts at each training step. The computational cost of this fresh generation is the fundamental systems-level challenge of OPD (Section 6.3).

Notation. Throughout this survey, we use lowercase p for token-level conditional distributions ($p_T(\cdot|x, y_{<t})$ for teacher, $p_\theta(\cdot|x, y_{<t})$ for student) and uppercase P for sequence-level distributions ($P_T(y|x)$, $P_\theta(y|x)$). $\mathcal{D}$ denotes the training dataset, p_{data} the data distribution, and $|V|$ the vocabulary size. D_{KL} and D_f denote KL divergence and general f-divergence respectively. In the RL formalization, π denotes policies and $R(x, y)$ the outcome reward. When discussing individual methods, we follow each paper's conventions where clarity demands.

2.1 Classical Knowledge Distillation

The original KD formulation (Hinton et al., 2015) trains the student to match the teacher's temperature-softened output distribution:

$$p_T^{(\tau)}(y|x) = \frac{\exp(z_y/\tau)}{\sum_j \exp(z_j/\tau)} \tag{2}$$

where z_y are the teacher's logits and $\tau > 1$ is a temperature parameter. When $\tau \to \infty$, the softened distribution approaches a uniform distribution over the vocabulary, revealing the teacher's "dark knowledge" about inter-class similarities that are hidden in the hard labels. The gradient of the distillation loss with respect to the student's logits $z_i^{\mathcal{S}}$ reveals why temperature matters:

$$\frac{\partial \mathcal{L}_{\text{KD}}}{\partial z_i^{\mathcal{S}}} = \frac{1}{\tau}(p_i^{\mathcal{S}} - p_i^{\mathcal{T}}) \tag{3}$$

In the high-temperature limit ($\tau \to \infty$), a Taylor expansion $\exp(z_i/\tau) \approx 1 + z_i/\tau$ shows that the gradient reduces to $\frac{1}{|V|}(z_i^{\mathcal{S}} - z_i^{\mathcal{T}})$, revealing that classical KD in this regime is equivalent to minimizing the mean squared error between raw logits. This forces the student to replicate the teacher's *entire* logit structure, transferring both primary predictive modes and nuanced sub-optimal probability masses (the "dark knowledge"). However, higher temperatures also amplify the contribution of non-peak tokens, effectively transferring the teacher's relational structure between vocabulary items. In practice, LLM distillation operates at moderate temperatures ($\tau \in [1, 3]$) or uses the teacher's natural distribution ($\tau = 1$), because the vocabulary is large enough that even untempered distributions contain rich inter-token structure.

For autoregressive LLMs, the token-level distillation loss factors as:

$$\mathcal{L}_{\text{Token-KD}} = \mathbb{E}_{x,y\sim\mathcal{D}}\left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t}))\right] \tag{4}$$

This is computationally tractable because it relies on static, pre-computed prefixes from the dataset, but it optimizes next-token accuracy while assuming an error-free history $y_{<t}$. Kim & Rush (2016b) extended this to Sequence-Level Knowledge Distillation (Seq-KD), which instead minimizes the global KL divergence over the entire sequence space:

$$\mathcal{L}_{\text{Seq-KD}} = D_{\text{KL}}(P_T(y|x) \parallel P_\theta(y|x)) = \sum_{y\in\mathcal{Y}} P_T(y|x) \log \frac{P_T(y|x)}{P_\theta(y|x)} \tag{5}$$

Since $|\mathcal{Y}| = |V|^T$ grows exponentially with length, exact computation is intractable. Seq-KD approximates $P_T(y|x)$ with a Dirac delta at the beam-search output $\hat{y} = \arg\max_y P_T(y|x)$, reducing to standard negative log-likelihood on teacher-generated sequences:

$$\mathcal{L}_{\text{Seq-KD}} \approx -\log P_\theta(\hat{y}|x) \tag{6}$$

This approximation works well when the teacher is highly confident but discards all information about the teacher's uncertainty and alternative modes, a critical limitation that motivates the softer distributional matching methods below.

Subsequent work (Zhong et al., 2024) revisited these foundations for modern LLMs, demonstrating that classical assumptions (small teacher-student gap, shared vocabulary, similar architectural inductive biases) frequently fail in the era of heterogeneous model families. This motivates the more general f-divergence framework below.

From classical KD to on-policy. The historical progression from Hinton's classical KD to modern OPD can be understood as a series of relaxations. Classical KD assumes: (1) shared vocabulary, (2) i.i.d. data, (3) static teacher, (4) off-policy training data. Seq-KD (Kim & Rush, 2016a) relaxes assumption (2) by operating at the sequence level but retains the static data assumption. GKD (Agarwal et al., 2024) relaxes assumption (4) by training on student-generated sequences. DSKD (Zhang et al., 2025b) relaxes assumption (1) through dual-space alignment. G-OPD (Yang et al., 2026c) relaxes all four simultaneously while adding RL-augmented objectives. Each relaxation addresses a specific failure mode that becomes more severe as model scale increases: vocabulary mismatch is irrelevant for same-family distillation but catastrophic for cross-family (relaxation 1), while exposure bias is negligible for short sequences but devastating for multi-step reasoning (relaxation 4). This progression is not merely historical but prescriptive: practitioners should identify which assumptions hold in their specific setting and select the minimally relaxed method, avoiding unnecessary complexity.

2.2 Off-Policy Exposure Bias

Standard Knowledge Distillation minimizes the KL divergence over states drawn from the dataset distribution $d_{\mathcal{D}}(s)$:

$$\mathcal{L}_{\text{Off-Policy}} = \mathbb{E}_{x,y\sim p_{\text{data}}}\left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t}))\right] \tag{7}$$

At inference, the student acts according to its own policy, inducing a different state visitation distribution $d_{\pi_\theta}(s)$. By the DAgger theorem (Ross et al., 2011), if a policy mimics an expert with per-step error ϵ under the training distribution, the expected total discrepancy over a trajectory of length T under the student's own distribution scales quadratically, bounded by $O(\epsilon T^2)$.

This quadratic compounding has severe practical consequences for reasoning tasks. A mathematical proof requiring 10 reasoning steps with per-step accuracy 95% yields only $(0.95)^{10} \approx 60\%$ trajectory-level correctness under off-policy training, because the student never learns to recover from its own intermediate errors. On-policy training exposes the student to its own error states during training, enabling it to learn correction strategies that reduce the effective per-step error rate when errors have already occurred.

Remark: The DAgger bound in LLMs. While the theoretical transition from $O(\epsilon T^2)$ to $O(\epsilon T)$ is appealing, applying the DAgger bound to LLMs requires nuance. The original theorem assumes an *interactive expert* providing optimal actions in any state. In white-box OPD, the "expert" provides a next-token distribution $p_T(y_t|\hat{y}_{<t})$ conditioned on the student's prefix $\hat{y}_{<t}$. If the student hallucinates a severely out-of-distribution prefix, the teacher's conditional distribution may itself become poorly calibrated, because the teacher was never trained on such inputs. Forcing the student to match this noisy distribution violates the core assumption of interactive imitation learning, potentially destabilizing training rather than recovering the $O(\epsilon T)$ bound. This observation provides the theoretical motivation for the adaptive trust mechanisms surveyed in Section 6.1 and explains why unconditional token-level matching is insufficient for robust OPD.

2.3 The Unified f-Divergence Framework

OPD resolves exposure bias by altering the training expectation to sample from the student's own rollouts, or a mixture policy π_{mix}. The generalized on-policy objective decouples the sampling trajectory from the local matching metric:

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{y \sim \pi_{\text{mix}}} \left[\sum_{t=1}^{|y|} \mathcal{D}_f \left(p_T(\cdot|x, y_{<t}),\ p_\theta(\cdot|x, y_{<t}) \right) \right] \tag{8}$$

Here, $\mathcal{D}_f$ represents a divergence from the f-divergence family (Wen et al., 2023). Formally, given two distributions P and Q, an f-divergence is defined as:

$$D_f(P \parallel Q) = \mathbb{E}_{y \sim Q} \left[f\left(\frac{P(y)}{Q(y)} \right) \right] \tag{9}$$

where $f : (0, \infty) \to \mathbb{R}$ is a convex generator with $f(1) = 0$. The choice of f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$:

- **Forward KL** ($f(u) = u \log u$): Mode-covering (zero-avoiding). The student must allocate probability mass wherever the teacher does, often bridging distinct teacher modes and hallucinating in the inter-mode space.
- **Reverse KL** ($f(u) = -\log u$): Mode-seeking (zero-forcing). The student collapses its mass onto the teacher's single highest peak, ignoring secondary acceptable outputs but guaranteeing high precision.
- **JSD** ($f(u) = u \log u - (u+1) \log \frac{u+1}{2}$): Symmetric, bounded, and smoothly interpolates between mode-covering and mode-seeking behavior.
- **α-divergence**: Parameterized family that continuously interpolates between Forward KL ($\alpha \to 1$) and Reverse KL ($\alpha \to 0$), enabling fine-grained control over the mode-seeking/covering tradeoff.

The practical implications of this taxonomy are profound. For tasks with a *unique* correct answer (mathematical proofs, code correctness), Reverse KL's mode-seeking behavior concentrates the student on the single optimal solution path. For tasks with *many* acceptable outputs (creative writing, open-ended QA), Forward KL's mode-covering behavior preserves output diversity. JSD offers a compromise but lacks theoretical optimality guarantees for either extreme. The key theoretical insight is that all these divergences are computable as expectations under the student's own policy (since $D_f(P_T \parallel P_\theta) = \mathbb{E}_{y \sim p_\theta}[f(p_T(y)/p_\theta(y))]$), which is precisely why they are amenable to on-policy optimization via policy gradient methods.

This framework subsumes all modern OPD objectives: the choice of f determines the geometric behavior, while the choice of π_{mix} controls the degree of on-policy exploration. We illustrate by mapping the three foundational methods into this space:

GKD (Agarwal et al., 2024) defines $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda)p_{\text{data}}$ as an explicit interpolation, and is agnostic to $\mathcal{D}_f$, empirically testing Forward KL, Reverse KL, and JSD. Setting $\lambda \to 1$ makes GKD purely on-policy, while $\lambda = 0$ reduces to standard off-policy KD. The λ parameter provides a smooth dial between the extremes: moderate values ($\lambda \approx 0.5$) provide the benefits of on-policy exposure while retaining the stability of grounded, dataset-derived prefixes. GKD's experiments establish that on-policy sampling ($\lambda = 1$) with JSD divergence yields the best downstream performance on summarization and translation, empirically validating the exposure bias hypothesis.

MiniLLM (Gu et al., 2024) selects Reverse KL as $\mathcal{D}_f = D_{\text{KL}}(p_\theta \parallel p_T)$ and employs $\pi_{\text{mix}} = (1-\alpha)p_\theta + \alpha p_T$ with $\alpha = 0.2$ for stability. Because Reverse KL places the student in both the expectation and the log-ratio, MiniLLM reformulates optimization via REINFORCE, treating $\log(p_T/p_\theta)$ as a per-step reward.

DistiLLM (Ko et al., 2024) maintains $\pi_{\text{mix}} = p_\theta$ (fully on-policy) but replaces the KL target from p_T to a skewed mixture $\tilde{p} = \alpha p_T + (1-\alpha)p_\theta$, computing $D_{\text{KL}}(p_T \parallel \tilde{p})$. This avoids zero-division instability when $p_\theta(y) \approx 0$, because $\tilde{p}(y) \geq \alpha p_T(y) > 0$ always, achieving tractability and stability without policy gradients.

This decomposition reveals that modern OPD is not a series of disparate heuristics but a systematic exploration of a design space governed by three interacting choices: trajectory sampling (π_{mix}), divergence generator (f), and argument ordering within the chosen divergence.

2.4 Distillation Scaling Laws

A critical gap in the field is the absence of scaling laws specific to distillation. While Hoffmann et al. (2022) established compute-optimal scaling for pre-training (the Chinchilla laws), and Busbridge et al. (2025) initiated the study of distillation-specific scaling, no comprehensive framework yet predicts how distillation quality scales with teacher size, student size, data volume, and on-policy rollout budget. This absence forces practitioners to rely on expensive trial-and-error. Preliminary evidence suggests that distillation scaling obeys qualitatively different laws from pre-training: the student benefits from teacher scale even after the teacher's marginal pre-training gains plateau, because the teacher's improved calibration (better next-token probability estimates) provides denser gradient signal per training step. Furthermore, on-policy rollout budget appears to exhibit a log-linear relationship with downstream quality once sufficient baseline capability is established, with roughly $2\times$ rollout budget yielding consistent 2–3% absolute improvements on reasoning benchmarks regardless of student scale. However, these observations remain empirical rather than predictive, and the theoretical frameworks developed in Section 7 provide only partial answers. A complete distillation scaling law, expressed as Quality $\propto N_T^\alpha \cdot N_S^\beta \cdot D^\gamma \cdot R^\delta$ where R is the rollout budget, remains an open problem (Section 9).

3 Landscape and Practitioner Guide

To navigate the fragmented OPD literature, we construct a functional landscape. Methods are classified not by arbitrary labels, but by the specific step in the engineering pipeline they optimize.

3.1 Method Landscape

The practitioner faces three sequential decisions: (1) what objective to optimize, (2) where the supervisory signal originates, and (3) how to stabilize the resulting training dynamics. Figure 1 illustrates this pipeline.

The three-stage pipeline is not merely organizational convenience: it reflects the actual engineering workflow. A practitioner first chooses an objective function (Forward KL? Reverse KL? Adaptive? RL-augmented?), then selects a signal source (white-box logits, black-box API, or self-generated), and finally addresses the training dynamics challenges

Objective Design (§4) ⟶ Signal Source (§5) ⟶ Training Dynamics (§6)		
Fixed/Adaptive/RL	White-Box/Black-Box/Self	Weighting/Curriculum/Efficiency

Figure 1: The functional landscape of On-Policy Distillation. The taxonomy reflects the practitioner's decision chain: first choose the objective, then identify the signal source, then stabilize the training loop.

that arise from the chosen combination. Some combinations are incompatible (e.g., Forward KL requires white-box logits, eliminating all black-box signal sources), which our decision tree (Section 3.3) makes explicit. Other combinations are synergistic: RL-augmented objectives (G-OPD) naturally pair with external feedback signal sources (verifiers, reward models), while fixed divergences pair most naturally with white-box sources where the full distribution is available for exact computation.

3.2 Method Comparison Table

Classification methodology. Each method is assigned to exactly one primary category based on its *core contribution dimension*: if a paper's novelty lies in the loss function (e.g., G-OPD's KL-constrained RL reformulation), it belongs to "Objective." If it introduces a novel signal architecture (e.g., DSKD's dual-space projection), it belongs to "Signal." If it solves a training stability or efficiency problem (e.g., TIP's token weighting), it belongs to "Dynamics." Methods that contribute to multiple dimensions are classified by their *most distinctive* contribution. This strict one-method-one-category rule prevents the overlap and confusion that plagued previous multi-dimensional taxonomies.

Table 1: Comparison of key OPD methods categorized by their primary contribution dimension.

Method	Category	Core Contribution	Signal Source	Loss Granularity
GKD (Agarwal et al., 2024)	Objective (Fixed)	DAgger for LLMs; interpolation	White-Box Logits	Token
DistiLLM (Ko et al., 2024)		Skewed KL mixtures	White-Box Logits	Token
MiniLLM (Gu et al., 2024)		Sequence-level Reverse KL via RL	White-Box Logits	Sequence
ToDi (Jung et al., 2025)	Objective (Adaptive)	Entropy-routed token divergence	White-Box Logits	Token
Entropy-Aware (Jin et al., 2026)		Entropy-based KL interpolation	White-Box Logits	Token
G-OPD (Yang et al., 2026c)	Objective (RL)	OPD equivalence to KL-RL	White-Box Logits	Sequence/Token
RLKD (Xu et al., 2025b)		KD augmented with reward model	Reward Model	Sequence
DSKD (Zhang et al., 2025b)	Signal (White-Box)	Cross-architecture alignment	White-Box Logits	Token
Delta-KD (Cao et al., 2025)		Base-to-Instruct delta isolation	White-Box Logits	Token
Lion (Jiang et al., 2023)	Signal (Black-Box)	Verbal feedback curriculum	Black-Box API	Sequence
GAD (Ye et al., 2025)		Adversarial distribution matching	Black-Box API	Sequence
OPSD (Zhao et al., 2026b)	Signal (Self)	Privileged Info (PI) conditioning	Self-Generated	Token
SPIN (Chen et al., 2024)		Self-play vs reference responses	Self-Generated	Sequence
TIP (Xu et al., 2026b)	Dynamics (Weight)	Token importance profiling via diff	White-Box Logits	Token
PACED (Xu et al., 2026a)		Beta-kernel curriculum pacing	White-Box Logits	Token

3.3 Practitioner Decision Tree

The following decision tree provides actionable guidance for selecting OPD methods based on access constraints, task requirements, and compute budgets. These recommendations are derived from the empirical patterns observed across the methods surveyed in Sections 4–6.

1. **Do you have full logit access to the teacher?**
   - *Yes (White-Box):* Use DistiLLM (Ko et al., 2024) for stable token-level matching on instruction following, or G-OPD (Yang et al., 2026c) if strong reasoning extrapolation is required. For adaptive divergences, ToDi (Jung et al., 2025) or Entropy-Aware (Jin et al., 2026) provide automatic switching. If teacher

and student use different tokenizers, deploy DSKD (Zhang et al., 2025b) or Cross-Tokenizer KD (Boizard et al., 2025).

- *No, API only (Black-Box):* Use OVD (Xiong et al., 2026) for verbal score distillation (simplest), GAD (Ye et al., 2025) for adversarial training (strongest signal), or DAIL (Mendes et al., 2026) for inverse-RL framing. If only text outputs are available and the student is small, LUFFY (Yan et al., 2025) with mixed-policy GRPO provides a strong baseline.
- *No teacher available (Self-Distillation):* If oracle answers are available, use Privileged Information methods: OPSD (Zhao et al., 2026b) for math reasoning, GATES (Stein et al., 2026) for open-domain QA. If no oracle is available, start with SD-ZERO (He et al., 2026) (requires only a binary verifier) or SSD (Zhang et al., 2026d) (requires nothing at all, but saturates quickly).

2. **What is the task profile?**
   - *Reasoning / Math:* Mode-seeking objectives are mandatory to prevent hallucination. Use Reverse KL (MiniLLM (Gu et al., 2024)) or RL-augmented KD (G-OPD, KDRL (Xu et al., 2025a)). Combine with curriculum pacing (PACED (Xu et al., 2026a)) for compute efficiency.
   - *Open-ended Generation:* Mode-covering objectives preserve diversity. Use Forward KL or adaptive entropy-aware divergences. Avoid aggressive Reverse KL, which collapses creative diversity.
   - *Instruction Following:* GKD (Agarwal et al., 2024) with $\lambda \geq 0.5$ provides the best quality-compute tradeoff. AlignDistil (Zhang et al., 2025a) adds preference alignment.
   - *Multi-step Agentic Tasks:* Requires sequence-level objectives with trajectory-aware credit assignment. SCoRe (Lyu et al., 2025) for error correction, RLAD (Zhang et al., 2026f) for trust-region stability.
3. **What are the compute limits?**
   - *Low compute (<500 GPU-hours):* Standard GKD, Fast OPD (Zhang et al., 2026a) (prefix truncation), or Lightning OPD (Wu et al., 2026) ($4\times$ cost reduction via offline caching).
   - *Medium compute:* Full on-policy training with adaptive divergences (ToDi, AKL (Wu et al., 2025)). Add PACED for efficient sample selection.
   - *High compute (>5000 GPU-hours):* Multi-stage pipelines: off-policy warmup $\rightarrow$ on-policy refinement $\rightarrow$ RL exploration. This is the recipe used by Qwen3 (Yang et al., 2025a) and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026).
4. **Stability concerns?**
   - *Length inflation:* Apply Stable-OPD (Luo et al., 2026) stabilization (reference divergence + rollout mixing).
   - *Flawed prefix trap:* Deploy TIP (Xu et al., 2026b) or SCOPE (Zheng et al., 2026) for teacher reliability filtering.
   - *Thinking-pattern mismatch:* Start with off-policy cold start (Li et al., 2026d), then transition to on-policy after initial distribution alignment.

4 Objective Functions and Optimization

The foundational algorithmic decision in OPD is the choice of objective function: what mathematical quantity does the student optimize at each training step? This choice governs gradient geometry, mode coverage, and ultimately the ceiling of achievable performance. The field has evolved through three distinct phases: fixed divergences that apply a single metric uniformly, adaptive divergences that modulate the metric per-token based on local distributional geometry, and RL-augmented objectives that break through the teacher's capability ceiling entirely. We trace this progression below, emphasizing the mathematical relationships that connect seemingly disparate methods.

4.1 Fixed Divergence Objectives

GKD (Agarwal et al., 2024) established the canonical OPD framework by adapting the DAgger algorithm (Ross et al., 2011) for autoregressive language models. It samples trajectories from a mixture policy $\pi_{\text{mix}} = \lambda p_\theta + (1-\lambda) p_{\text{data}}$ and applies standard f-divergences (Forward KL, Reverse KL, or JSD) at each token position. The mixture coefficient λ interpolates between fully off-policy ($\lambda = 0$) and fully on-policy ($\lambda = 1$), providing a single knob for exposure bias mitigation. On instruction-following and summarization benchmarks, even moderate on-policy mixing ($\lambda \geq 0.5$) consistently outperforms pure off-policy training, validating the distributional alignment hypothesis.

However, pure KL divergences exhibit numerical instability during on-policy exploration. When the student generates tokens to which the teacher assigns near-zero probability, Forward KL produces unbounded gradients. Conversely, Reverse KL ignores these regions entirely, creating blind spots in the loss landscape. DistiLLM (Ko et al., 2024) addresses this through *Skewed KL* (SKL), constructing a smoothed target $\tilde{p} = \alpha p_T + (1-\alpha) p_\theta$:

$$\mathcal{L}_{\text{SKL}} = \mathbb{E}_{y \sim p_\theta} \left[\sum_{t=1}^{|y|} D_{\text{KL}}\big(p_T(\cdot|y_{<t}) \parallel \alpha p_T(\cdot|y_{<t}) + (1-\alpha) p_\theta(\cdot|y_{<t})\big) \right] \tag{10}$$

Skewing guarantees the target density is bounded below by $(1-\alpha)p_\theta$, eliminating gradient explosion without requiring RL policy gradients. DistiLLM further introduces Skewed Reverse KL (SRKL) for mode-seeking applications, creating a symmetric pair of stabilized divergences. Its successor, DistiLLM-2 (Ko et al., 2025), recognizes that teacher-generated and student-generated data carry fundamentally different learning signals and applies *asymmetric* objectives: Forward SKL on teacher-generated data (where mode-covering teaches the student to explore the teacher's full distribution) and Reverse SRKL on student-generated data (where mode-seeking teaches the student to sharpen its own best responses). This source-aware asymmetry yields consistent improvements over symmetric single-divergence baselines across instruction following and summarization tasks, empirically validating the theoretical insight that the optimal divergence is data-source-dependent.

The fixed-divergence family faces a fundamental limitation: the choice between Forward KL (mode-covering) and Reverse KL (mode-seeking) is a global decision applied uniformly across all token positions. Yet the teacher's distributional structure varies dramatically within a single sequence. At a mathematical operator token, the teacher concentrates probability mass on one or two correct symbols. At a conjunction or filler word, dozens of tokens share comparable probability. Applying a single divergence to both positions necessarily compromises: Forward KL wastes gradient signal on high-entropy positions where any token is acceptable, while Reverse KL discards useful coverage information at low-entropy positions where multiple valid continuations exist.

Sequence-level objectives offer a complementary perspective. MiniLLM (Gu et al., 2024) elevates Reverse KL to the full sequence level, where the student p_θ appears inside the sampling expectation. The central challenge is computing the gradient of an expectation over p_θ when p_θ itself is parameterized by θ. Starting from $D_{\text{KL}}(p_\theta \parallel p_T) = \mathbb{E}_{y \sim p_\theta}[\log p_\theta(y|x) - \log p_T(y|x)]$ and applying the log-derivative trick:

$$\nabla_\theta \mathcal{L}_{\text{MiniLLM}} = \mathbb{E}_{y \sim p_\theta} \left[\Big(\log p_\theta(y|x) - \log p_T(y|x) \Big) \nabla_\theta \log p_\theta(y|x) \right] \tag{11}$$

This reveals that minimizing sequence-level Reverse KL is mathematically equivalent to policy gradient RL with reward $r(x,y) = \log p_T(y|x) - \log p_\theta(y|x)$. MiniLLM decomposes this into per-step rewards r_t and future returns $R_t = \sum_{t'=t}^{|y|} r_{t'}$, enabling variance-reducing baselines:

$$\nabla_\theta \mathcal{L}_{\text{MiniLLM}} = -\mathbb{E}_{y \sim p_\theta} \left[\sum_{t=1}^{|y|} (R_t - b)\, \nabla_\theta \log p_\theta(y_t|y_{<t}) \right] \tag{12}$$

where b is a learned baseline. This places optimization squarely in the RL framework, with the teacher's log-probability serving as a dense reward signal. The connection to policy optimization explains MiniLLM's strong performance on reasoning tasks where sequence-level coherence matters more than token-level accuracy. However, REINFORCE estimation in massive combinatorial output spaces demands substantially more training iterations and sophisticated baseline subtraction to converge.

KETCHUP (Fan et al., 2025) refines this approach by replacing the single-sample REINFORCE estimator with a K-step truncated return, reducing gradient variance while preserving the sequence-level objective. The constrained KD framework of Zimmer et al. (2025) augments the sequence-level Reverse KL with a hard KL constraint implemented via state-augmented CMDP, ensuring that the student's divergence from the teacher remains bounded even during aggressive exploration.

The token-level vs. sequence-level tradeoff. The choice between token-level and sequence-level objectives represents a fundamental bias-variance tradeoff. Token-level methods (GKD, DistiLLM) provide $|V|$-dimensional gradient signals at every position, yielding low variance but potentially biased gradients when the teacher's token-level distribution is poorly calibrated (the flawed prefix trap of Section 6.1). Sequence-level methods (MiniLLM, KETCHUP) provide unbiased gradients through the policy gradient theorem but suffer from high variance due to single-sample REINFORCE estimation over exponentially large sequence spaces. The practical resolution is task-dependent: for instruction following where token-level teacher quality is high, GKD-style methods dominate. For reasoning tasks where sequence-level coherence is paramount, MiniLLM-style methods are preferred despite their variance, and KETCHUP's truncated returns offer an intermediate point on this spectrum.

4.2 Adaptive Divergence Objectives

The limitations of fixed divergences motivate *adaptive* methods that select or interpolate divergences based on local distributional geometry. The core insight is that the optimal divergence at each token position depends on the teacher's local confidence: when the teacher is certain (low entropy), mode-seeking Reverse KL preserves precision, and when the teacher is uncertain (high entropy), mode-covering Forward KL preserves diversity.

ToDi (Jung et al., 2025) operationalizes this by routing each token through either Forward KL or Reverse KL based on the teacher's entropy $H(p_T(\cdot|y_{<t}))$:

$$\mathcal{L}_{\text{ToDi}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} 1[H_t < \tau] \cdot D_{\text{KL}}(p_\theta \parallel p_T) + 1[H_t \geq \tau] \cdot D_{\text{KL}}(p_T \parallel p_\theta)\right] \tag{13}$$

where τ is a learned or fixed entropy threshold. Entropy-Aware OPD (Jin et al., 2026) replaces the hard routing with a continuous interpolation coefficient $\alpha_t \propto H_t$, yielding a smooth blend:

$$\mathcal{L}_{\text{EA}} = \mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} (1-\alpha_t) D_{\text{KL}}(p_\theta \parallel p_T) + \alpha_t D_{\text{KL}}(p_T \parallel p_\theta)\right] \tag{14}$$

Both methods consistently outperform any fixed divergence on math reasoning (MATH, GSM8K) and open-ended generation (AlpacaEval), confirming that the optimal divergence is indeed position-dependent.

AKL (Wu et al., 2025) takes a more aggressive approach, dynamically interpolating between Forward KL and Reverse KL based on a per-token confidence metric derived from the teacher-student probability ratio. Unlike entropy-based routing, which depends only on the teacher's distributional shape, AKL considers the *relative* geometry of teacher and student, adapting more aggressively at positions where the gap is largest.

These adaptive methods admit a natural interpretation through information geometry. The Fisher information metric on the statistical manifold of autoregressive distributions varies across token positions. At reasoning-critical positions, the manifold is highly curved (the teacher concentrates probability on few tokens), making Reverse KL's mode-seeking behavior locally appropriate because it follows the manifold's natural gradient. At generation-flexible positions, the manifold is nearly flat (many synonyms are equally valid), and

Forward KL's mode-covering behavior preserves useful entropy. Adaptive divergences implicitly navigate this varying curvature by selecting the locally optimal metric, which explains their consistent empirical superiority over any fixed choice.

The practical implication is clear: the field has moved decisively from "which divergence?" to "which divergence *where*?" The evidence from ToDi, AKL, and Entropy-Aware OPD converges on a single recommendation: no fixed divergence is ever optimal, and the simplest adaptive routing (even a hard entropy threshold) consistently outperforms the best-tuned fixed alternative.

4.3 RL-Augmented Objectives

Divergence matching, whether fixed or adaptive, intrinsically limits the student to the teacher's capability ceiling: perfect optimization yields a student that matches (but never exceeds) the teacher's distribution. RL-augmented objectives break this ceiling by introducing an external performance signal that can steer the student toward solutions the teacher would never produce.

OPD as KL-constrained RL. G-OPD (Yang et al., 2026c) establishes a formal equivalence between standard OPD and dense KL-constrained reinforcement learning. It formulates distillation as maximizing a token-level reward derived from teacher logits, penalized by divergence from a reference policy:

$$\max_{\theta} \mathbb{E}_{y \sim p_\theta} \left[\sum_{t=1}^{|y|} \alpha \log p_T(y_t | y_{<t}) - D_{\mathrm{KL}}\big(p_\theta(\cdot|y_{<t}) \parallel p_{\mathrm{ref}}(\cdot|y_{<t})\big) \right] \tag{15}$$

When $\alpha = 1$, this reduces to standard Reverse KL distillation. When $\alpha > 1$, the objective forces the student to *extrapolate* beyond the teacher's probability mass, enabling a phenomenon documented as *Reward Extrapolation*: the student discovers novel, structurally sound reasoning paths in regions where the teacher's generation probability $p_T(y|x)$ is low but the outcome reward remains high. On competition-level math, a 4B student trained with G-OPD surpasses a 30B teacher, demonstrating that OPD need not be a lossy compression step.

This equivalence has a deeper theoretical consequence: it bridges the distillation and RL communities, providing a shared analytical language. Any advance in KL-constrained RL (better trust regions, adaptive penalty coefficients, variance reduction) immediately transfers to OPD, and vice versa.

Gradient decomposition and the KD+RL complementarity. The unified KD+RL framework of Li et al. (2025) decomposes the hybrid gradient into two orthogonal components: a dense KD gradient for token-level imitation and a Monte Carlo RL gradient for reward maximization. The KD component dampens the high variance inherent in policy gradient estimation, while the RL component prevents the student from collapsing onto teacher modes that are suboptimal for the downstream task. REOPOLD (Ko et al., 2026) operationalizes this insight, interpreting on-policy distillation as policy optimization where the teacher-student log-likelihood ratio acts as a token reward, and stabilizing training through mixture-based reward clipping (preventing over-trust of extreme teacher signals), entropy-based dynamic sampling (selectively leveraging teacher feedback at high-uncertainty tokens), and a unified exploration-to-refinement strategy. Empirically, REOPOLD achieves 6.7–12× greater sample efficiency than recent RL approaches and enables a 7B student to match a 32B teacher in visual reasoning with ∼3.3× inference speedup.

Dense KD + sparse reward. RLKD (Xu et al., 2025b) exemplifies the integration of dense teacher supervision with sparse outcome feedback. It introduces a Generative Structure Reward Model (GSRM) that scores reasoning paths for structural coherence, combining token-level KL regularization with trajectory-level reward:

$$\max_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim p_\theta(\cdot|x)} \left[R(x, y) - \beta D_{\mathrm{KL}}\big(p_\theta(\cdot|x) \parallel p_T(\cdot|x)\big) \right] \tag{16}$$

The mathematical structure explains why the combination works: the KL term provides analytically computable dense gradients, while the reward term contributes Monte Carlo policy gradients for outcome-driven exploration. Empirically, RLKD demonstrates that structure-aware rewards can markedly improve sample efficiency, surpassing standard SFT-RL pipelines even when trained on only 0.1% of the data.

Joint vs. sequential optimization. A persistent engineering challenge is *how* to combine KD and RL within a training pipeline. Sequential approaches (distill-then-RL or RL-then-distill) suffer from catastrophic interference: the RL stage erases distilled knowledge through policy drift, while the KD stage suppresses novel solutions discovered through exploration. KDRL (Xu et al., 2025a) operationalizes joint optimization by adding an on-policy KL regularizer between student and teacher *during* RL training. RLAD (Zhang et al., 2026f) refines this with Trust Region Ratio Distillation (TRRD), replacing the standard KL regularizer with a PPO-style likelihood-ratio objective anchored to a teacher-old-policy mixture. This selective strategy follows the teacher only when its signal improves the policy update and ignores it otherwise, a principled solution to the problem that blind KL minimization can drag the student toward suboptimal trajectories on problems the teacher itself struggles with.

When logit access is insufficient to capture global intent, several methods map the problem onto Direct Preference Optimization (DPO). AlignDistil (Zhang et al., 2025a) constructs synthetic preference pairs from student rollouts scored by the teacher, applying both standard and reverse DPO objectives. OVD (Xiong et al., 2026) queries the teacher for pairwise preferences over student-generated responses, avoiding continuous probability extraction entirely. These preference-based objectives share a fundamental connection to token-level KL methods: under DPO's closed-form solution, the optimal policy satisfies $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x)\exp(r(y,x)/\beta)$, and substituting the teacher's log-probability as the implicit reward yields the geometric mixture distribution targeted by GKD with λ-mixing. The practical implication is that DPO-style methods are most useful when constructing preference pairs is cheaper than computing full teacher logits.

A distinct failure mode arises from the *uniformity* of all methods above: when the student makes a single error in step 3 of a 10-step proof, wholesale imitation retrains all 10 steps, diluting the corrective gradient. SuperCorrect (Yang et al., 2025b) introduces hierarchical thought templates with cross-model DPO for error localization. SCoRe (Lyu et al., 2025) pushes targeting to its logical extreme, correcting *only the earliest error* in the student's trajectory, concentrating all supervision at the single point of maximum leverage. This exploits a structural property of reasoning chains: errors cascade, so correcting the root cause eliminates all downstream consequences simultaneously.

From signal matching to environment reconstruction. The most radical extension asks whether the student can recover the teacher's *underlying optimization landscape*. $\mathcal{X}$-KD (Cai & Yuan, 2026) adopts the AVRIL framework to jointly model the teacher's latent reward function and perform policy distillation, encouraging the student to optimize in a reconstructed version of the teacher's training environment rather than simply mimicking its outputs. TSD-KD (Kim & Baek, 2026) combines indirect distillation (student generates candidates that the teacher re-ranks via preference feedback) with direct distillation (KL matching applied selectively to high-entropy tokens where teacher-student gaps are largest). These methods represent the current frontier: moving from matching the teacher's *behavior* to understanding its *incentives*.

The progression across objective types reveals a fundamental insight. Fixed divergences provide stable, well-understood optimization but are limited by the teacher's capability ceiling. Adaptive divergences improve efficiency by matching the local geometry of the loss landscape. RL-augmented objectives break through the ceiling entirely but introduce variance and instability. The optimal choice depends on the specific application: for instruction following where teacher quality is high, fixed or adaptive divergences suffice. For reasoning tasks where the student must exceed the teacher, RL augmentation is essential. For tasks where the teacher's training process is accessible, environment reconstruction offers the most complete knowledge transfer.

5 Signal Source and Teacher Architecture

The second major decision in the OPD pipeline concerns the origin and nature of the supervisory signal. Given a fixed objective function (Section 4), the practitioner must determine *where the teacher signal comes from*: full logit access (white-box), API-only text outputs (black-box), or the model's own internal asymmetries (self-distillation). Each signal source imposes distinct constraints on the applicable objectives and achievable performance, creating a structured design space that this section maps comprehensively.

The evolution of signal sources mirrors the broader trend toward accessibility and self-sufficiency. Early OPD methods (GKD, DistiLLM) assumed full white-box access to model internals, a requirement that limits distillation to same-organization deployments. The rise of proprietary API-only models (GPT-4, Claude) necessitated black-box methods that can extract knowledge from text outputs alone. Most recently, self-distillation methods eliminate the external teacher entirely, enabling continuous improvement without access to any stronger model. This progression, from full access through limited access to no external access, represents increasing autonomy at the cost of signal density, creating a fundamental trade-off that practitioners must navigate.

Complex reasoning is where OPD proves most indispensable and where off-policy methods encounter their hardest limits. The reason is structural: reasoning is *path-dependent*. A proof that begins "assume for contradiction..." requires entirely different subsequent tokens than one beginning "we proceed by induction..." If the student has never generated the inductive approach during training, it cannot recover at inference time when that path would be optimal. Off-policy training over a fixed dataset of teacher reasoning traces can only cover a finite set of solution paths, while the student needs to navigate its own unique landscape of partial solutions, dead ends, and recovery strategies. This insight motivates the on-policy formulation for reasoning transfer: the student generates its own intermediate reasoning steps $r^S \sim P_\theta(\cdot|x)$, and the teacher evaluates these student-generated chains rather than dictating its own:

$$\mathcal{L}_{\text{CoT-OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D},\, r^S\sim P_\theta(\cdot|x)}\left[\sum_{t=1}^{|r^S|} D_{\text{KL}}\Big(P_T(\cdot|x, r^S_{<t}) \parallel P_\theta(\cdot|x, r^S_{<t})\Big)\right] \tag{17}$$

The Reverse KL direction is mode-seeking by design: rather than forcing the student to spread probability across all teacher-approved continuations (which would blur distinct reasoning strategies), it reinforces the student's own high-probability paths that the teacher validates. This self-consistent learning dynamic, where the student practices its own reasoning style under expert supervision, is why on-policy CoT distillation has become the de facto standard for instilling multi-step reasoning into smaller models (Hsieh et al., 2023).

5.1 White-Box Logit Supervision

White-box access provides the densest possible signal: the full probability distribution across $|V|$ tokens at every decoding step. This exposes the teacher's implicit decision boundaries, including the relative probabilities among incorrect tokens (Hinton's "dark knowledge"), which carry information about similarity structure that binary labels cannot capture.

Standard logit distillation. The majority of methods surveyed in Section 4 assume white-box access. GKD, DistiLLM, ToDi, G-OPD, and their variants all require computing $p_T(\cdot|x, y_{<t})$ at each student-generated token position. The computational cost is dominated by the teacher forward pass, which must be executed once per student-generated token (or once per sequence for cached implementations).

Token-level adaptive supervision. Within the white-box setting, several methods refine *which* tokens receive supervision. SelecTKD (Huang et al., 2025) applies confidence-based filtering, discarding loss gradients from positions where the teacher's top-1 probability falls below a threshold. AdaSwitch (Peng et al., 2025) dynamically switches between exploration (student generates freely) and guidance (teacher provides dense supervision) based on the student's cumulative error on the current prefix, balancing the exploration benefit of on-policy generation against the stability of teacher guidance. Token-Adaptive KD (Xie et al., 2025) computes per-token importance weights using the gradient magnitude of the KL loss, concentrating compute on tokens where the teacher-student gap is largest. This approach is complementary to entropy-based routing (ToDi, Entropy-Aware): while the latter selects *which divergence* to use at each position, Token-Adaptive KD selects *how much gradient signal* to propagate, providing a second dimension of per-token adaptation.

The temporal scope of white-box distillation is not limited to post-training. MiniPLM (Gu et al., 2025) extends on-policy distillation to the *pre-training* stage, showing that distilling during pre-training yields stronger foundations for downstream fine-tuning. This represents a qualitative shift: rather than being a post-hoc compression step, OPD becomes an integral part of the model creation pipeline. Gemma 2 (Gemma Team et al., 2024) validates this approach at industrial scale, embedding KD directly into the pre-training loop for its 27B→9B→2B model cascade. The success of pre-training distillation raises a deeper question: if OPD can improve models during their initial training, should it be viewed as a separate post-processing step at all, or as an intrinsic component of the training objective alongside next-token prediction?

A critical challenge arises when teacher and student use different tokenizers or vocabulary sizes. Direct KL divergence over mismatched vocabularies is undefined. DSKD (Zhang et al., 2025b) solves this through *dual-space* knowledge distillation, introducing projectors that map between teacher and student representation spaces:

$$\mathcal{L}_{\text{DSKD}} = D_{\text{KL}}(P_{T\to S} \parallel P_S) + D_{\text{KL}}(P_{S\to T} \parallel P_T) \tag{18}$$

where $P_{T\to S}$ denotes the teacher's hidden states projected into the student's space and decoded through the student's prediction head. An exact token alignment algorithm handles vocabulary mismatches by identifying identical tokens across tokenizers. The key insight is that vocabulary tokens with similar semantic embeddings should receive similar probability mass, even when the tokenization schemes differ. This enables distillation across fundamentally different architectures (e.g., Llama teacher to Qwen student) that share no vocabulary overlap.

Cross-Tokenizer KD (Boizard et al., 2025) takes a complementary approach based on optimal transport, computing an alignment matrix between teacher and student token sequences that minimizes the earth mover's distance in latent space:

$$\mathcal{L}_{\text{CrossTok}} = \inf_{\pi \in \Pi(P_S, P_T)} \mathbb{E}_{(z_S, z_T)\sim\pi} \left[\| W_{S\to T} z_S - z_T \|_2^2 \right] \tag{19}$$

The distinction between hidden-state alignment (DSKD) and vocabulary-space alignment (Cross-Tokenizer KD) highlights a fundamental design trade-off. Hidden-state alignment preserves richer representational structure but requires maintaining two projectors during training. Vocabulary-space alignment is more lightweight but operates at a level where information has already been compressed through the vocabulary bottleneck. In either case, the alignment maps must be *co-trained* with the student, because the student's representation space shifts during on-policy training as its policy evolves, a form of representation drift absent in off-policy settings. Concrete Score Matching (Kim et al., 2026b) provides yet another solution, distilling via a continuous score function that bypasses the discrete vocabulary mismatch entirely. Together, these methods ensure that OPD is architecture-agnostic, not limited to same-family distillation. This has immediate practical implications: organizations can now distill from any available teacher (e.g., Llama-based) into any student architecture (e.g., Qwen-based) without the historically restrictive requirement of shared vocabulary, enabling a marketplace of heterogeneous teacher-student pairs that was previously infeasible.

Beyond selecting *which tokens* to supervise, a complementary question is *what information* the teacher should provide at each token. The answer is not "everything": not all content in the teacher's logits is equally useful. Delta-KD (Cao et al., 2025) introduces the principle of *signal isolation*, distilling only the differential shift between a base model and its instruction-tuned variant. Since the base model's distribution captures generic language modeling knowledge the student already possesses through pre-training, distilling it is redundant at best and harmful at worst (it wastes gradient budget on information the student has already internalized). By extracting only the instruction-following delta, Delta-KD provides a cleaner supervisory signal that transfers alignment without overwriting the student's foundation.

A related but distinct challenge is teacher *overconfidence*: when the teacher assigns extreme probability to tokens outside the student's reachable space, it creates optimization targets that are simultaneously precise and useless. Two complementary solutions address this from opposite directions. Veto (Jang et al., 2026) (Adaptive Target Reformulation) refines the teacher's output *post-hoc*, constructing a geometric bridge in logit space with a parameter α that serves as both an Adaptive Gradient Veto (suppressing pathological gradients on low-confidence tokens) and a Decisiveness Knob (balancing performance with diversity), dynamically reformulating the teacher's target distribution based on the student's current capability. PromptKD (Kim et al., 2024) takes the opposite approach, adapting the teacher *pre-hoc* by prepending learnable prompt tokens (adding only 0.0007% of the teacher's parameters) that cause the teacher to produce "student-friendly" distributions more accessible to the student's limited capacity. The distinction is instructive: post-hoc methods (Veto, Delta-KD) operate on the supervisory signal after it is produced, while pre-hoc methods (PromptKD) modify the teacher's generation process itself. Both are effective, and their composition is orthogonal.

TAID (Shing et al., 2025) introduces yet another dimension of adaptation: temporal interpolation of the target itself. It constructs a time-dependent intermediate distribution $P_t = (1 - \lambda_t)P_\theta + \lambda_t P_T$ that starts close to the student and progressively approaches the full teacher distribution as training proceeds. This temporal curriculum over the target (rather than over the input prompts as in PACED) provides a complementary axis of control: PACED asks "which problems to train on," while TAID asks "how much of the teacher's knowledge to reveal at each stage." The progression from PACED (problem-level curriculum) through AdaSwitch (token-level timing) to TAID (target-level interpolation) and PromptKD (teacher-level adaptation) traces an important design spectrum: curriculum can operate at four distinct granularities (prompt, token, target, teacher), all of which address the same underlying bottleneck of keeping the difficulty gap between teacher signal and student capability in the productive learning range. Their orthogonality means they compose naturally, and the most sophisticated future systems will likely employ all four levels simultaneously.

5.2 Black-Box and API-Constrained Distillation

When only API access is available, the student observes the teacher's top-1 text output (or at best, top-k log-probabilities) rather than the full distribution. This information bottleneck eliminates token-level divergence matching and forces methods to operate at the sequence level. Despite this limitation, black-box methods have proven surprisingly effective, suggesting that much of the teacher's knowledge is recoverable from text-only observations.

Distilling Step-by-Step. Hsieh et al. (2023) demonstrated an influential early approach: extracting not just the teacher's final answers but its intermediate reasoning steps (rationales) as additional supervision. The student is trained with a multi-task objective combining answer prediction and rationale generation, enabling it to outperform the teacher on downstream tasks with as little as 1,000$\times$ less data. This work established the principle that *how* the teacher reasons is often more valuable than *what* it concludes, a principle that pervades modern OPD.

Adversarial distribution matching. GAD (Ye et al., 2025) reformulates the distillation problem as a generative adversarial game:

$$\max_G \min_D \; V(G, D) = \mathbb{E}_{(x,y^T)\sim\mathcal{T}} \left[-\log \sigma\Big(D(y^T) - D(G(x)) \Big) \right] \tag{20}$$

A discriminator D (initialized from the student with a scalar prediction head) distinguishes student rollouts from teacher API outputs via a Bradley-Terry preference model, while the generator G maximizes V via REINFORCE using the discriminator score as an evolving on-policy reward signal. Unlike white-box methods that match full distributions, GAD extracts knowledge through a scalar quality signal, trading distributional fidelity for API compatibility. It inherits the training instability of adversarial objectives, as the discriminator can overfit to stylistic cues rather than semantic quality, but achieves consistent gains over SFT baselines.

Verbal feedback and curriculum. Lion (Jiang et al., 2023) exploits the teacher's capacity as a natural language critic through a three-stage adversarial loop: (1) *imitation*, where the student is fine-tuned on teacher-generated responses, (2) *discrimination*, where the teacher identifies instructions on which the student still falls short, and (3) *generation*, where the teacher creates harder instructions targeting the student's identified weaknesses. This closed-loop curriculum progressively narrows the teacher-student gap without requiring any gradient information from the teacher, making it applicable to any instruction-following API. Lion-13B achieves competitive performance on BIG-Bench Hard and AGIEval using only 70K training examples.

Didactic-to-constructive learning. DAIL (Mendes et al., 2026) bridges the gap between high-quality expert solutions that are out-of-distribution for the student and the student's actual reasoning space. The core challenge is that simply fine-tuning on expert solutions yields poor transfer because the student never learns to navigate its own reasoning space. DAIL applies a two-step process: first transforming expert solutions into detailed, in-distribution reasoning traces through inverse reinforcement learning (inferring the implicit reward function guiding the teacher's generation choices from observed text trajectories), then applying a contrastive objective to focus learning on expert methodologies rather than surface patterns. Using fewer than 1,000 expert solutions, DAIL achieves 10–25% pass@k gains. This result underscores a principle that recurs throughout this survey: the quality of the supervision signal matters less than its *alignment* with the student's current distribution.

Distribution-Aligned Sequence Distillation (DASD) (Yan et al., 2026) critically reexamines the dominant practice of SFT on teacher-generated reasoning traces, identifying three fundamental limitations: (1) inadequate representation of the teacher's sequence-level distribution from single best-of-n samples, (2) misalignment between teacher output distribution and student capacity, and (3) exposure bias from teacher-forced training. Its enhanced distribution-aligned pipeline addresses all three by generating multiple diverse teacher traces per prompt and aligning the student's generation distribution with the teacher's through sequence-level matching, achieving state-of-the-art performance using only 448K training samples. DDT (Zhang et al., 2026c) complements DASD's empirical findings with a theoretical framework explaining why RL generalizes better than SFT for reasoning tasks: the key factor is RL's on-policy data generation, which keeps training aligned with the model's evolving distribution rather than a static snapshot.

Verbal score distillation. OVD (Xiong et al., 2026) replaces token-level probability matching entirely with trajectory matching using discrete verbal scores (0–9) from teacher models. This eliminates the requirement for token-level alignment between vocabularies, dramatically reduces memory consumption by avoiding logit storage, and delivers up to +12.9% absolute EM improvement on web QA and +25.7% on math benchmarks with a single rollout sample. The surprising effectiveness of such coarse supervision demonstrates that on-policy exploration itself provides a substantial fraction of the learning signal, with the teacher's role being more to *select among student trajectories* than to *correct individual tokens*.

The progression from adversarial (GAD) through verbal (Lion, OVD) to preference-based (AlignDistil, ORPO-Distill) methods reflects a fundamental spectrum of *signal richness versus access requirements*. Adversarial methods recover the densest proxy for the teacher's distribution but require the most complex training dynamics (discriminator co-training, generator-discriminator instability). Score-based methods are simpler to train but lose distributional information, recovering only a scalar quality signal. Preference-based methods occupy a principled middle ground, recovering relative quality without requiring calibrated absolute scores, and their connection to DPO-style objectives provides well-understood convergence guarantees. The practical implication is that practitioners should select the signal type that matches their API access tier: full logit access $\rightarrow$ KL methods (Section 4), token-level logits unavailable $\rightarrow$ adversarial or score-based, ranking-only access $\rightarrow$ preference-based.

Preference-based formulations. ORPO-Distill (Singh et al., 2025) maps black-box distillation onto preference optimization, generating candidate responses from the student and querying the teacher API for pairwise rankings. The mixed-policy strategy, combining on-policy student traces with off-policy teacher traces, outperforms both pure off-policy and pure on-policy alternatives, suggesting that the contrastive structure of preference learning (learning what *not* to do alongside what to do) provides complementary signal to the positive-only feedback of score-based methods. ThinkTuning (RRV et al., 2025) occupies an intermediate position between black-box distillation and RL: rather than providing scores after the fact, a same-size teacher reviews the student's incorrect answers *during* the GRPO training loop and provides short corrective hints (rather than complete solutions). This interleaving of corrective feedback episodes with RL training acts as an implicit privileged signal that shapes policy gradient updates, and is particularly effective for base models that lack strong prior reasoning behaviors, achieving +3.85% over zero-shot baselines across benchmarks (+2.08% on MATH-500, +2.23% on AIME, +3.99% on GPQA-Diamond over vanilla GRPO).

Off-policy knowledge integration. A related challenge is incorporating knowledge from off-policy sources (e.g., pre-existing teacher-generated datasets) into an on-policy framework. LUFFY (Yan et al., 2025) addresses this by extending GRPO to a mixed-policy objective that incorporates off-policy reasoning traces alongside the student's on-policy rollouts. The critical design challenge is preventing *superficial imitation*: without correction, the student tends to copy high-probability tokens from off-policy traces while ignoring low-probability but critical reasoning steps. LUFFY's policy shaping via regularized importance sampling reweights gradients to emphasize learning from unfamiliar yet effective actions, achieving +6.4 average points over standard RLVR.

5.3 Self-Distillation: Privileged Information

Self-distillation eliminates the need for an external teacher entirely. Instead, the model exploits asymmetries *within itself* to construct a training signal. The first family of self-distillation methods creates this asymmetry through *privileged information* (PI): training-time access to context that is unavailable at inference. The central insight is that conditioning the same model on additional context produces a strictly stronger policy that can supervise the unconditioned version, turning a single network into both teacher and student.

Ground-truth as PI. OPSD (Zhao et al., 2026b) constructs the teacher policy by conditioning on both the problem x and the ground-truth answer $y^\star$, while the student conditions only on x. The student generates on-policy rollouts, and the teacher provides dense token-level supervision:

$$\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(x,y^\star)\sim\mathcal{S}}\mathbb{E}_{\hat{y}\sim p_\theta(\cdot|x)}\left[\sum_{n=1}^{|\hat{y}|} D\big(p_\theta(\cdot|x,y^\star,\hat{y}_{<n}) \parallel p_\theta(\cdot|x,\hat{y}_{<n})\big)\right] \tag{21}$$

On competition-level math benchmarks (AIME 2024/2025, HMMT), OPSD matches or exceeds GRPO at 4B and 8B scales while using only a single rollout per problem versus GRPO's eight, achieving a significant computational advantage. At 1.7B scale, however, the gains over GRPO are minimal, indicating that sufficient model capacity is necessary for the PI mechanism to provide useful supervision. The teacher's ground-truth conditioning must produce meaningfully different token distributions from the student, which requires enough parameters to represent the conditional shift.

A complementary limitation arises when all rollouts for a given prompt fail (the "cliff prompt" problem), causing the policy gradient to vanish entirely. This failure mode directly parallels PACED's finding (Section 6.1) that gradient SNR vanishes when student pass rate approaches zero, but with an important difference: PACED addresses this through curriculum selection (avoiding problematic prompts), while HDPO (Ding, 2026) addresses it through signal augmentation. HDPO augments GRPO with a JSD-based self-distillation term on ground-truth-conditioned rollouts, ensuring non-zero gradients even at the boundary of the student's competence, directly patching the cliff-prompt vulnerability without restricting the problem distribution.

Relaxing the ground-truth requirement. Ground-truth labels are not always available. GATES (Stein et al., 2026) demonstrates that *any* training-time context can serve as PI by having a single model act as tutor (conditioned on a source document) and student (answering from the question alone). Because source documents do not guarantee correct answers, GATES introduces a consensus-based gating mechanism that suppresses the gradient when tutor uncertainty is high, preventing the model from reinforcing unreliable self-supervision. Penaloza et al. (2026) generalize the OPSD and GATES paradigms into a unified theoretical framework, formalizing the conditions under which privileged self-distillation provably improves over standard training.

Behavioral PI: context distillation. The PI paradigm extends beyond factual ground truth to behavioral instructions. OPCD (Ye et al., 2026b) treats system prompts, exemplars, and retrieval context as privileged information, enabling a model to internalize behaviors that would otherwise require expensive in-context demonstrations at inference time. OPSDL (Anonymous, 2026a) extends this to long-context settings: applying on-policy self-distillation to enhance long-context capabilities without relying on high-quality supervision data or sparse sequence-level rewards, enabling stable optimization for context length scaling beyond what standard post-training methods achieve. OEL (Ye et al., 2026a) extends to continuous post-deployment learning, using interaction outcomes as dynamic PI that updates the model's teacher policy as it accumulates experience, closing a loop that OPCD opened.

Reasoning compression via PI. CRISP (Sang et al., 2026) demonstrates that a model prompted to "be concise" can serve as a teacher for its own verbose version, reducing chain-of-thought token count by 57–59% on MATH-500 while *improving* accuracy by 9–16 percentage points. This reveals that much of the verbosity in distilled reasoning models is a trainable inefficiency, not a necessary feature of competent reasoning.

5.4 Self-Distillation: Self-Play

A more radical question is whether a model can improve using nothing but its own rollouts, with no PI, no external feedback, and no additional context. The answer is a qualified yes, subject to important saturation limitations.

Self-play as a two-player game. SPIN (Chen et al., 2024) casts self-improvement as an adversarial game between successive checkpoints, training the updated model to distinguish its previous iteration's generations from human-written responses:

$$\mathcal{L}_{\text{SPIN}} = \mathbb{E}_{x, y \sim p_{\text{data}}, y' \sim p_{\theta_t}} \left[\ell \left(\lambda \left(\log \frac{p_{\theta_{t+1}}(y|x)}{p_{\theta_t}(y|x)} - \log \frac{p_{\theta_{t+1}}(y'|x)}{p_{\theta_t}(y'|x)} \right) \right) \right] \tag{22}$$

SPIN provides a theoretical convergence guarantee: the global optimum of $\mathcal{L}_{\text{SPIN}}$ is achieved if and only if $p_{\theta_{t+1}} = p_{\text{data}}$, i.e., the student's distribution exactly matches the reference distribution. Crucially, at this fixed point, the loss is zero and the gradient vanishes, so the iterative process is *stable*: once the student reaches the target distribution, further iterations cannot destabilize it. The game-theoretic interpretation is that p_{data} is the unique Nash equilibrium of the two-player game, and SPIN's iterative updates correspond to fictitious play converging to this equilibrium. Starting from Zephyr-7B-SFT, it improves MT-Bench from 5.94 to 6.78 over 3 iterations, with diminishing returns at each round. However, this convergence guarantee comes with a fundamental limitation: SPIN converges to p_{data}, not to the optimal policy. If the reference data is sub-optimal (as human-written responses often are), SPIN will faithfully reproduce those sub-optimalities, revealing that pure self-play recycles existing knowledge rather than generating new knowledge.

IRIS (Anonymous, 2026b) provides a unifying framework for this family by replacing the implicit KL divergence in SPIN with an interpolative Rényi divergence parameterized by $\alpha \in (0, \infty)$, recovering SPIN ($\alpha \to 1$, KL regime), SPACE ($\alpha \to 0.5$, Jensen-Shannon regime), and SPIF ($\alpha \to 2$, χ^2-regularized regime) as special cases. The Rényi parameter controls the exploration-exploitation tradeoff: lower α encourages broader distributional coverage (exploration) while higher α concentrates updates on the most likely improvements (exploitation). This unification reveals that the various self-play methods in the literature differ only in their implicit divergence choice, not in their fundamental mechanism, and suggests that α-tuning can be used as a meta-parameter for adapting self-play behavior to task requirements.

This motivates the external grounding mechanisms of Section 5.5.

Temporal self-supervision. SDFT (Shenfeld et al., 2026) positions on-policy self-distillation as a mechanism for forgetting resistance via implicit KL regularization against the model's own prior checkpoint, converting the model's optimization trajectory into a regularization resource for continual learning.

Self-play for efficiency. MTP via self-distillation (Kirchenbauer et al., 2026) converts a pre-trained autoregressive model into a multi-token predictor by training an auxiliary prediction head on the model's own next-token predictions, achieving $> 3\times$ faster decoding at $< 5\%$ accuracy drop. This demonstrates that self-distillation can target *architectural* improvements (inference speed) rather than *behavioral* improvements (task accuracy), expanding the scope of what OPD can optimize.

On-Policy SFT (Zhao et al., 2026a) provides a surprising simplification: by filtering self-generated responses for correctness (via a verifier) and conciseness (via length truncation at the median successful length), standard SFT on this filtered on-policy data matches or exceeds full RL training. The key insight is that the combination of on-policy generation (which naturally produces diverse solution strategies) with outcome filtering (which selects the best strategies) implicitly performs a form of evolutionary optimization that requires no gradient through the reward function. This "generate-filter-train" paradigm dramatically reduces implementation complexity compared to full RL-based distillation while retaining the core benefit of on-policy data: distributional alignment.

Self-Distilled RLVR (Yang et al., 2026a) decomposes the gradient update into two components: self-distillation determines the *magnitude* of per-token policy updates (how much to change each token's probability), while environmental RLVR feedback anchors the *direction* (which tokens should increase vs. decrease). This decomposition combines the stability of distillation with the grounding of external verification. At just 200 training steps, RLSD surpasses GRPO trained for 400 steps on Qwen3-VL-8B-Instruct, demonstrating $2\times$ sample efficiency from the complementary signal structure.

Minimalist self-distillation. SSD (Zhang et al., 2026d) requires no RL, no verifier, no external teacher, and no code execution. It samples solutions at training-time temperature, fine-tunes via standard SFT, and reshapes token distributions in a context-dependent manner. On LiveCodeBench v6, SSD improves Qwen3-30B-Instruct from 42.4% to 55.3%, revealing that when the base model is sufficiently capable, even the simplest self-play yields substantial gains. The critical requirement is not a sophisticated algorithm but sufficient *diversity* in the model's sampling distribution to produce high-quality variants that can serve as training targets.

However, this conclusion also reveals the fundamental boundary of pure self-play: when diversity is achieved through temperature sampling alone, the model's improvement is ultimately bounded by the information already present in its pre-training distribution. Bridging this boundary requires either connecting the model to external reality (Section 5.5) or internally generating privileged information through the self-play process itself (π-Play below).

extbfUnifying PI and self-play. π-Play (Zhang et al., 2026e) unifies privileged information and self-play within a single multi-agent self-evolution framework. The key observation is that self-play naturally produces a byproduct: during task generation, the examiner agent constructs a question construction path (QCP) that provides high-quality PI at no additional cost. Three co-evolving agents (examiner π_ϕ^E, teacher π_ψ^T conditioned on QCP, student π_θ^S without it) are jointly optimized through alternating updates. As the student strengthens, the examiner generates harder tasks, and the teacher's behavior is softly constrained to track the student via a KL penalty, establishing a self-adjusting curriculum that requires no external data. On Qwen3 (4B, 8B), data-free π-Play surpasses supervised search agents (Search-R1 by 5–15%) and improves efficiency by 2–3$\times$ over conventional self-play (Dr.Zero), with the largest gains on multi-hop benchmarks where the teacher's token-level credit assignment provides the most leverage.

π-Play demonstrates that the self-play saturation problem can be mitigated *without* external feedback by internally generating privileged information through the self-play process itself. This mechanism is orthogonal to the external-verification strategies of Section 5.5, suggesting that the two approaches could be combined for even stronger self-improvement.

5.5 Self-Distillation: External Feedback

The self-play methods above require no external input but face an inherent ceiling: without ground-truth or verifier access, the model cannot distinguish confident-but-wrong outputs from genuinely correct ones, leading to the saturation problem analyzed in Section 7.2. External feedback methods retain the self-distillation architecture (no separate teacher model) while injecting non-differentiable truth signals, typically code execution, symbolic verification, or outcome rewards, that anchor the self-play loop and prevent it from collapsing onto self-reinforcing errors.

Pure self-play is inherently bounded by the information already present in the model's pre-training distribution. External signals, such as compiler outputs, test results, and textual critiques, ground self-distillation in objective reality and bridge the gap to reinforcement learning.

From binary rewards to dense supervision. SD-ZERO (He et al., 2026) converts sparse verifier signals into dense self-supervision through a dual-role architecture: the Generator produces on-policy responses, and the Reviser, conditioned on the Generator's output and its binary correctness signal, produces an improved version. The Generator then distills from the Reviser's token-level distribution, converting binary feedback into dense self-supervision. On Qwen3-4B-Instruct, SD-ZERO achieves 68.3% on AIME 2024, outperforming GRPO (62.5%), demonstrating that self-revision can be more effective than standard RL for converting sparse rewards into learning signal. SDPO (Hübotter et al., 2026) extends this beyond binary feedback to *structured textual* feedback: runtime errors, failing unit tests, and LLM judge evaluations construct fine-grained credit assignment that attributes success or failure to specific reasoning steps rather than entire trajectories. RL from Text Feedback (RLTF) (Song et al., 2026) pushes further with free-form natural language critiques from an automated judge, training the model's single-turn policy to match its own feedback-conditioned second-turn generations.

The progression from SD-ZERO's binary signal through SDPO's structured errors to RLTF's free-form critiques traces a clear trajectory: as external feedback becomes richer, the self-distillation objective becomes more precisely targeted, enabling the model to correct specific failure modes rather than globally adjusting its policy. Sample-Routed Policy Optimization (SRPO) (Li et al., 2026b) demonstrates that these approaches can be composed by routing: correct samples follow GRPO's reinforcement path (reward-weighted likelihood maximization) while failed samples follow SDPO's targeted logit-level correction. The insight is that correct and incorrect samples carry distinct types of learning signal, and treating both identically wastes the distinctive information each carries. SRPO raises the average by 3.4% over GRPO and 6.3% over SDPO alone, confirming that the optimal external-feedback strategy is a routing function matching each sample to its most informative objective.

Environmental grounding. RLTF and related methods use compiler outputs, unit test results, and symbolic math verifiers as external feedback that shatters self-reinforcing echo chambers. Even if the model becomes highly confident in a flawed derivation, the verifier assigns $R = 0$, providing an unambiguous correction signal that pure self-play cannot generate.

Saturation analysis. Kim et al. (2026a) trace self-distillation degradation in mathematical reasoning to the suppression of *epistemic verbalization*: the model's expression of uncertainty during reasoning. When self-distillation shortens reasoning traces, it disproportionately removes hedging phrases and uncertainty markers that serve as implicit "attention flags" for difficult sub-problems. The result is shorter but less calibrated reasoning, where the model confidently commits to flawed steps. This connects to the teacher uncertainty challenge in Section 7: an overconfident student produces unreliable self-play targets, and the saturation problem ultimately requires either external grounding or the internally generated PI mechanism of π-Play to overcome.

The progression across signal sources, from white-box logits through black-box APIs to self-distillation, mirrors a fundamental trade-off between signal density and autonomy. White-box methods provide the richest supervision ($|V|$-dimensional distributions at every token) but require same-organization deployment and co-location of teacher and student. Black-box methods sacrifice distributional information but operate across organizational boundaries, with recent methods (GAD, OVD) recovering surprisingly effective proxy signals. Self-distillation eliminates external dependencies entirely but is bounded by the model's pre-existing capabilities unless augmented by external verification or internally generated PI. This continuum is not merely descriptive: it predicts where future methods will have the most impact. The white-box extreme is well-explored (GKD, DistiLLM, DSKD are mature), while the self-distillation extreme still lacks principled convergence guarantees for complex reasoning. The greatest innovation potential lies at the boundaries: methods like ThinkTuning that blur the line between black-box feedback and RL, or methods like π-Play that blur the line between self-play and externally grounded verification.

6 Training Dynamics and Efficiency

Given an objective function (Section 4) and a signal source (Section 5), the final engineering decision is *how to stabilize and accelerate the on-policy training loop*. On-policy generation introduces unique optimization challenges absent from standard supervised fine-tuning: non-stationary data distributions (the student's policy shifts during training, making older rollouts stale), gradient signal-to-noise ratio (SNR) collapse on hard prompts (where all rollouts fail and no useful gradient emerges), and the computational overhead of autoregressive student rollouts followed by teacher scoring (typically 4–5$\times$ more expensive than off-policy training).

These challenges are not independent. SNR collapse on hard prompts wastes compute, motivating curriculum strategies that avoid such prompts. Non-stationarity invalidates cached teacher signals, motivating efficient online scoring. And the compute overhead of autoregressive generation creates a systems-level bottleneck that requires architectural solutions. The methods in this section address these challenges through three complementary mechanisms.

6.1 Token and Sample Weighting

Not all teacher supervision is equally reliable in the on-policy setting. The core problem is the *flawed prefix trap*: when a student generates an erroneous token early in a sequence, all subsequent teacher predictions are conditioned on this out-of-distribution prefix. Because the teacher has never been trained on such prefixes, its conditional distribution $p_T(\cdot|x, \hat{y}^{\text{bad}}_{<t})$ becomes noisy and unreliable. Naively matching this corrupted signal propagates errors rather than correcting them.

Teacher reliability filtering. TIP (Xu et al., 2026b) (Token Importance Profiling) provides a principled framework for token weighting by organizing importance along two axes: student entropy h_t and teacher-student divergence δ_t. Crossing these axes yields four quadrants with qualitatively distinct learning roles: high-entropy, high-divergence tokens (Q1) carry the dominant corrective signal, high-entropy, low-divergence tokens (Q2) indicate agreement on uncertainty, low-entropy, high-divergence tokens (Q3) represent *overconfident errors* where the student is certain but wrong, and low-entropy, low-divergence tokens (Q4) are genuinely mastered. The key theoretical result shows that Q3 tokens are structurally invisible to any entropy-only weighting scheme, because no non-decreasing function of entropy with $f(0) = 0$ can distinguish them from Q4. TIP's parameter-free Soft-OR score $s_t = \hat{h}_t + \hat{\delta}_t - \hat{h}_t \cdot \hat{\delta}_t$ recovers Q3 coverage while preserving Q1/Q2 sensitivity, achieving consistent improvements at 50% token retention across three model families with capacity gaps from 2× to 9×.

SCOPE (Zheng et al., 2026) elevates this analysis from tokens to *rollouts*, revealing the "flawed prefix trap": when the student produces an erroneous prefix, the teacher's conditional distribution becomes poorly calibrated because it has never been trained on such inputs. SCOPE's dual-path architecture routes rollouts by correctness: incorrect trajectories receive teacher-perplexity-weighted KL distillation (down-weighting unreliable guidance), while correct trajectories receive student-perplexity-weighted MLE (concentrating reinforcement at the capability boundary). This addresses diversity collapse, the counterintuitive phenomenon where Pass@1 improves at the expense of Pass@k, yielding +5.5% over standard OPD.

SelecTKD (Huang et al., 2025) applies confidence-based filtering at the token level, discarding loss gradients from positions where the teacher's top-1 probability falls below a threshold. The reasoning is that low teacher confidence indicates either genuine ambiguity (where any student token is acceptable) or distributional confusion (where the teacher cannot provide useful supervision), and in neither case does the gradient carry reliable information.

AdaSwitch (Peng et al., 2025) introduces a dynamic switching mechanism between pure exploration and guided distillation. Unlike static filtering that simply attenuates unreliable signals, AdaSwitch maintains a running estimate of the student's cumulative prefix quality and switches between on-policy exploration (when the prefix is reasonable) and off-policy guidance (when the prefix has drifted too far). This single-switch design preserves semantic coherence (avoiding the jarring token-level alternation of prior mixing methods) while ensuring high-quality supervision at critical junctures, and is computationally simpler than continuous weighting while empirically achieving similar stability.

Sample-level weighting. Beyond token-level filtering, several methods weight entire samples based on their informativeness. The flawed prefix trap manifests most severely on *hard* prompts where the student's pass rate approaches zero. On these prompts, nearly all rollouts contain early catastrophic errors, and the resulting teacher signals are dominated by noise. Methods that uniformly sample prompts waste most of their gradient budget on uninformative hard prompts or trivially correct easy prompts, with the productive learning concentrated on prompts at the boundary of the student's competence.

Failure-mode analysis. Fu et al. (2026) provide a systematic empirical analysis of why sampled-token OPD is fragile in long-horizon settings. They identify three failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on student-generated prefixes that deviate from the teacher's training distribution, and (3) distortions caused by tokenizer or special-token mismatch. Their proposed fix, teacher top-K local support matching via truncated Reverse KL with top-p rollout sampling and special-token masking, yields more stable optimization across math reasoning and agentic training.

The token and sample weighting methods above operate *reactively*: they attenuate the loss signal *after* the student has already generated a potentially uninformative rollout. A more proactive approach asks: can we select prompts *before* generation to maximize the expected information gain per rollout? This motivates the curriculum methods below.

6.2 Curriculum and Difficulty Adaptation

The sample weighting problem motivates a more principled solution: *curriculum design* that actively selects prompts matching the student's current competence level.

Competence-boundary sampling. PACED (Xu et al., 2026a) models prompt difficulty dynamically using a Beta-kernel distribution centered on the student's current pass rate. The curriculum progressively shifts toward harder prompts as the student improves, but critically avoids prompts where the pass rate is near zero (where gradient SNR vanishes) or near one (where the student has already mastered the content). This "frontier sampling" strategy concentrates the gradient budget on the narrow band of prompts where learning is most efficient.

The mathematical justification connects to the gradient SNR analysis: for a prompt with pass rate p, the expected gradient magnitude scales as $p(1-p)$, which is maximized at $p = 0.5$ and vanishes at both extremes. PACED's Beta-kernel sampling directly optimizes for this quantity, providing a theoretically grounded alternative to uniform prompt sampling. Concretely, PACED maintains a running estimate of per-prompt difficulty $\hat{p}_i$ (computed from the last m rollouts) and samples each prompt with probability proportional to $\text{Beta}(\hat{p}_i; \alpha, \beta)$ where (α, β) are hyperparameters controlling the width of the competence frontier. Early in training, wide kernels cast a broad net. As training converges, narrow kernels focus on the diminishing frontier of unsolved problems. This reduces wasteful rollouts on already-mastered problems by 40–60% compared to uniform sampling, translating directly into compute savings proportional to the curriculum efficiency.

This SNR analysis has a deeper implication that extends beyond curriculum design: it explains *why* on-policy training naturally outperforms off-policy on reasoning tasks without explicit curriculum engineering. In off-policy settings, the problem difficulty distribution is fixed by the dataset. As the student improves, an increasing fraction of training data falls into the "too easy" zone where gradients are negligible, leading to diminishing returns. On-policy training, by contrast, automatically adjusts the effective difficulty because the student's rollouts reflect its current capabilities, naturally generating sequences at the boundary of its competence. PACED makes this automatic adjustment explicit and optimal, but even naive on-policy training captures much of the benefit simply by sampling from p_θ rather than p_{data}.

Self-adjusting curricula. π-Play's multi-agent framework (Section 5.4) provides an alternative curriculum mechanism: as the student strengthens, the examiner agent automatically generates harder tasks, creating a self-adjusting difficulty ladder that requires no external difficulty estimation. PAINT (Tan & Hong, 2026) revisits the privileged on-policy self-distillation setting through a contextual re-scoring lens, introducing overlap-adaptive partial-solution masking that controls how much verified solution context is revealed during re-scoring, combined with sparse energy interpolation to stabilize self-distillation gradients. PRISM (Wang et al., 2026b) addresses the multimodal setting, proposing black-box on-policy distillation as a pre-alignment stage before RLVR: an adversarial MoE discriminator distills from stronger model demonstrations (Gemini 3 Flash) into Qwen3-VL without requiring white-box access, providing a structured warm-up that substantially improves subsequent RL training stability.

Off-policy cold start. Li et al. (2026d) identify that OPD can fail catastrophically when the student's initial policy is too far from the teacher's distribution (the "thinking-pattern mismatch" problem). Their proposed solution is an off-policy cold-start phase: a brief period of standard SFT on teacher-generated data that reduces the pattern gap before switching to on-policy training. This two-phase approach, off-policy warmup followed by on-policy refinement, has become the standard industrial pattern (Section 8).

Retaining knowledge during on-policy training. A complementary concern is that on-policy training can cause *catastrophic forgetting* of capabilities the student acquired during pre-training. Chen et al. (2025) ("Retaining by Doing") demonstrate that on-policy data generation itself serves as an implicit rehearsal mechanism: by generating sequences from its own distribution, the student naturally revisits and reinforces its existing capabilities while learning new ones. This insight suggests that the on-policy generation step provides a dual benefit: it both exposes the student to its own error states (enabling correction) and rehearses its existing capabilities (preventing forgetting), offering a theoretical explanation for why on-policy methods empirically suffer less from catastrophic forgetting than sequential fine-tuning approaches.

Semantic bootstrapping. Semantic Soft Bootstrapping (Mitra & Ulukus, 2025) provides a different perspective on curriculum design: rather than adapting the *difficulty* of prompts, it adapts the *semantic granularity* of supervision. The method provides the model with both correct and incorrect rollouts as in-context exemplars that shape the next generation, progressively refining from coarse semantic alignment to fine-grained token-level matching. This achieves +10.6% on MATH-500 over GRPO, demonstrating that in-context self-supervision can complement gradient-based optimization.

The hybrid SFT+OPD pipeline. The off-policy cold start (Li et al. (2026d)) combined with on-policy refinement has become the de facto industrial recipe. Empirically, the optimal transition point from off-policy to on-policy training occurs when the student's generation distribution achieves sufficient overlap with the teacher's (typically measured by a token-level agreement rate >60%). Transitioning too early wastes on-policy compute on uninformative rollouts (the student is too weak for meaningful self-generation). Transitioning too late leaves the student stuck in the off-policy ceiling. Qwen3 (Yang et al., 2025a) uses approximately 20% off-policy warmup followed by 80% on-policy refinement. DeepSeek-R1 (DeepSeek-AI et al., 2025) uses 100% off-policy (no transition), which works at extreme teacher capability but leaves performance on the table for reasoning tasks where the student could exceed the teacher through on-policy exploration.

6.3 Compute Optimization

OPD requires generating student rollouts *and* scoring them with the teacher at every training step, creating a compute overhead of 3–8× compared to off-policy SFT. Several methods target this bottleneck.

The cost breakdown for a typical white-box OPD step consists of:

1. **Student rollout** ($\sim$40% of step time): autoregressive generation of $B \times K$ completions (batch size B, K rollouts per prompt), memory-bound due to KV cache growth.
2. **Teacher scoring** ($\sim$35% of step time): forward pass through the teacher to obtain full-vocabulary logits on student-generated sequences, compute-bound.
3. **Student update** ($\sim$25% of step time): standard backward pass computing gradients of the distillation loss, dominated by optimizer state memory.

Each component admits independent optimization, and the methods below attack different bottlenecks.

Prefix truncation. Fast OPD (Zhang et al., 2026a) observes that exposure bias is concentrated in the early tokens of a sequence, where errors compound most severely. It truncates on-policy rollouts to a prefix of length k and reverts to off-policy KD for the remainder, achieving comparable quality at substantially reduced cost. The optimal truncation point k depends on the task: for mathematical reasoning, where errors cascade rapidly, short prefixes suffice. For open-ended generation, longer on-policy segments are needed.

Offline teacher caching. Lightning OPD (Wu et al., 2026) decouples student optimization from teacher inference entirely by precomputing teacher distributions into an offline memory bank and asynchronously refreshing them. When teacher distributions are *consistent* (i.e., the teacher's output depends primarily on the prompt and only weakly on the specific student prefix), the cached distributions remain valid even as the student's policy evolves. Lightning OPD demonstrates that offline distillation under teacher consistency can match online OPD at 4× lower cost, significantly lowering the barrier for academic research.

These two methods reveal a deeper structural insight: the "on-policy" requirement is not binary but admits degrees of approximation. Fast OPD shows that partial rollouts preserve most of the distributional alignment benefit. Lightning OPD shows that stale teacher signals are acceptable when the student's policy drift is bounded. Together, they define a *fidelity–efficiency frontier* that practitioners can navigate based on their compute budget, with full on-policy training at one extreme (maximum alignment, maximum cost) and fully offline SFT at the other (minimum cost, exposure bias ceiling).

Speculative knowledge distillation. Speculative KD (Xu et al., 2025c) adapts speculative decoding to the distillation setting: the student generates candidate tokens that the teacher *verifies and selectively replaces*, amortizing the teacher's inference cost across multiple student tokens per teacher forward pass. DistillSpec (Zhou et al., 2023) inverts this: instead of using speculation to speed up training, it uses on-policy KD to improve the draft model for speculative decoding at *inference* time. The expected speedup:

$$E[S] = \frac{\mathbb{E}[K_{\text{accepted}}] + 1}{1 + c} \quad \text{where } c = \frac{\text{cost(student forward)}}{\text{cost(teacher forward)}} \tag{23}$$

This reveals a virtuous cycle: better distillation produces better draft models, which enable faster speculative decoding, which in turn enables more efficient distillation, progressively reducing the computational barrier.

GPU memory management. Beyond FLOPs, the most severe constraint for white-box OPD is GPU memory. Holding a 70B teacher alongside a 7B student requires the teacher's weights ($\sim$140 GB in BF16), the student's weights and optimizer states ($\sim$84 GB), and the full-vocabulary logits tensor $[B, T, |V|]$ for distillation. Peak memory can easily exceed an 8×80 GB H100 node. Practitioners alleviate this via: (1) *teacher quantization*, serving the teacher in FP8 or INT4 for a 2–4$\times$ memory reduction at negligible precision loss, (2) *logit offloading*, immediately streaming logits to CPU RAM or retaining only the top-k entries (since 99% of the probability mass is typically concentrated in fewer than 100 tokens), and (3) *aggressive gradient checkpointing* to minimize student activation memory during the backward pass. Combining these techniques is essential for viable white-box OPD on standard clusters.

Concrete cost example. To ground these formulas, consider distilling a 70B teacher into a 7B student on 8$\times$H100 GPUs. Off-policy over 1B tokens: the teacher generates the dataset offline ($\sim$200 GPU-hours), the student trains for $\sim$100 GPU-hours, totaling $\sim$300 GPU-hours. On-policy over 1B tokens: each step requires (1) student generation ($\sim$3$\times$ forward-pass cost due to autoregressive decoding), (2) teacher scoring (one 70B forward pass), and (3) student backward pass, yielding $\sim$1,200–1,500 GPU-hours, a 4–5$\times$ overhead. However, for reasoning tasks, on-policy methods consistently outperform off-policy SFT, and the off-policy performance ceiling is strictly lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This "ceiling gap" widens with task complexity, providing the strongest cost-benefit justification for on-policy methods.

7 Understanding OPD: Theory, Failure Modes, and Cost

The preceding sections addressed the *how* of on-policy distillation: what to optimize (Section 4), where the signal comes from (Section 5), and how to stabilize training (Section 6). This section turns to the equally important *why* and *when*: under what conditions does OPD succeed, how does it fail, what theoretical frameworks unify the diverse methods surveyed above, and when should practitioners invest in OPD versus simpler off-policy pipelines?

Crucially, these four dimensions of understanding are not independent: they form a cumulative argument. Identifying *when* OPD works (Section 7.1) reveals the preconditions that the theoretical framework must explain. Cataloging *how* it fails (Section 7.2) motivates the specific design choices, adaptive divergences, curriculum weighting, and hybrid RL objectives, whose theoretical grounding Section 7.3 provides. Both the success conditions and the failure modes then inform the practical cost-benefit analysis in Section 7.4, which distills the theoretical insights into engineering guidelines. Answering these questions is essential for moving OPD from an empirical art toward a principled engineering discipline with predictable outcomes. The analysis below draws on both theoretical frameworks (imitation learning bounds, information geometry, RL convergence theory) and systematic empirical investigations that isolate individual design choices.

7.1 Success Conditions

Li et al. (2026d) provide the most systematic investigation of OPD training dynamics, identifying two necessary conditions for effective distillation.

Two conditions jointly govern whether OPD yields improvement. First, the student and teacher must share compatible reasoning patterns (operationalized as high overlap in their top-k token distributions): even a superior teacher fails to transfer knowledge if its thinking style is fundamentally mismatched with the student's (e.g., a non-thinking teacher distilling into a thinking student), because low initial distributional overlap cannot be recovered through training alone. Second, the teacher must provide genuinely new capabilities beyond what the student has already acquired. When both models are trained on the same data and recipe, they converge to similar distributions at their respective scales, leaving the teacher with little transferable signal, as reverse distillation experiments show that same-family 1.5B and 7B teachers become distributionally indistinguishable from the student's perspective.

At the token level, successful OPD is characterized by progressive alignment on *high-probability overlap tokens*, a small shared token set concentrating 97–99% of the probability mass. When these conditions fail, recovery strategies include off-policy cold start (SFT initialization that reduces the pattern gap) and teacher-aligned prompt selection. Critically, the quality of OPD's dense supervision degrades systematically with trajectory depth, with instability originating at later tokens and propagating backward, raising fundamental questions about OPD's applicability to long-horizon reasoning.

These findings point toward a general principle: OPD's benefit scales with the *exploitable gap* between teacher and student. When this gap is too small (same-recipe models), the teacher offers no new information. When too large (thinking-pattern mismatch), the student cannot absorb the supervision. The productive regime lies in between, and identifying it is the central diagnostic challenge.

Diagnostic checklist. Based on the surveyed literature, we distill four pre-training diagnostics that practitioners should evaluate before committing to an OPD pipeline:

1. *Token overlap ratio* (>60%): Compute top-k overlap between teacher and student on a held-out set. If below 60%, apply off-policy cold start first.
2. *Pass rate on target prompts* (>5%): If the student cannot solve any problems in the target domain, OPD gradients will vanish. Use curriculum (PACED) or easier warm-up tasks.
3. *Teacher calibration*: Verify that the teacher's confidence correlates with correctness on student-generated prefixes. Poorly calibrated teachers inject noise.
4. *Length stability*: Monitor output length during early training. Monotonic growth indicates the KL gradient asymmetry identified by Luo et al. (2026). Apply Stable-OPD corrections preemptively.

7.2 Failure Modes

Complementing the success conditions, a growing body of work catalogs the specific ways OPD can fail *even when structural preconditions hold*, as artifacts of the on-policy training dynamics themselves. Understanding these failure modes is essential for debugging OPD pipelines in practice, and each mode has a corresponding mitigation strategy developed in the methods sections above.

We organize failure modes by their *root cause* rather than their symptom, as multiple failure modes can produce similar surface-level degradation (e.g., accuracy collapse) through different mechanisms:

The flawed prefix trap. As detailed in Section 6.1, when the student generates an erroneous prefix, the teacher's conditional distribution over subsequent tokens is poorly calibrated because the teacher was never trained on such out-of-distribution inputs. Fu et al. (2026) formally characterize three specific failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on out-of-distribution student prefixes, and (3) tokenizer mismatch distortions. The theoretical analysis reveals a fundamental bias-variance tradeoff: token-level OPD is biased relative to sequence-level Reverse KL but enjoys tighter worst-case variance bounds.

Two subtler failure modes compound the prefix trap. *Gradient SNR collapse* occurs when the student's pass rate on a prompt approaches zero: all rollouts contain early errors, teacher signals are dominated by noise, and the gradient SNR vanishes, explaining why hard prompts yield no learning signal in standard OPD and motivating PACED's competence-boundary curriculum (Section 6.2). *Epistemic suppression* (Kim et al., 2026a) manifests in self-distillation: when the process shortens reasoning traces, it disproportionately removes hedging phrases and uncertainty markers that serve as implicit attention flags for difficult sub-problems, producing shorter but less calibrated reasoning where the model confidently commits to flawed steps. This calibration degradation connects directly to the teacher uncertainty challenge: an overconfident student produces unreliable self-play targets, creating a vicious cycle of increasing certainty and decreasing reliability.

Self-play saturation (the Ouroboros problem). In self-distillation settings, the student optimizes against a target sharing its own inductive biases and architectural limitations, causing the hypothesis space to progressively collapse. If the model discovers a syntactic "hack" or a highly confident but flawed reasoning heuristic, no external signal penalizes it. The model self-reinforces this flawed trajectory, driving $p_\theta(y_{\text{flawed}}|x) \to 1$. Once entirely certain of its own hallucinations, gradients vanish, exploration ceases, and the policy is trapped. This is distinct from exposure bias (which arises from train-test distribution mismatch in off-policy settings): saturation arises from the *absence* of distributional diversity in fully on-policy self-play, and is mathematically analogous to mode collapse in generative adversarial networks.

The precision-recall tradeoff. Cha & Cho (2025) formalize a fundamental tension in distillation: as the student concentrates mass on high-quality outputs (improving precision), it necessarily reduces coverage of the teacher's full output distribution (reducing recall). In the OPD setting, this manifests as the "diversity collapse" problem: aggressive Reverse KL distillation produces a student that scores highly on Pass@1 (precision) but catastrophically on Pass@k (recall), because the student has collapsed onto a single reasoning strategy per prompt. On-policy generation can *shift* this tradeoff by allowing the student to explore its own distribution rather than being constrained to teacher-demonstrated paths, potentially improving recall without sacrificing precision. Temperature-scaled sampling during on-policy generation, as used in GKD and G-OPD, provides a direct control knob: higher sampling temperatures encourage distributional exploration (increasing recall) at the cost of more noisy gradients (potentially reducing precision). This tradeoff is modulated by a single entropy-controlling parameter, providing practitioners with a principled axis for tuning distillation behavior to downstream requirements.

The calibration-capability gap. CaOPD (Zhang et al., 2026b) reveals a "Scaling Law of Miscalibration": while OPD effectively improves task accuracy, it systematically traps models in severe overconfidence. The root cause is an information mismatch: teacher supervision is formed under privileged context available during training (full reasoning traces, multiple samples), whereas the deployed model must report confidence using only its deployment-time context. By decoupling capability improvement from calibration degradation, CaOPD demonstrates that the distilled model becomes more capable but less aware of its own uncertainty boundaries. This finding has important implications for deployment: a distilled model that achieves higher benchmark scores but cannot reliably indicate when it might be wrong is arguably less safe than a weaker but better-calibrated baseline. The "illusion of certainty" connects to the saturation analysis above: both indicate that OPD can produce students that lose metacognitive capabilities (uncertainty awareness, epistemic verbalization) even as task-level performance improves.

7.3 Unified Theoretical Perspectives

The success conditions and failure modes above raise a natural question: can they be explained by a single theoretical framework? Several recent works demonstrate that they share common mathematical roots.

The unified f-divergence framework of Wen et al. (2023) subsumes Forward KL, Reverse KL, JSD, and α-divergences as special cases parameterized by a single convexity function f. Since all these divergences can be expressed as expectations under the student's on-policy distribution, the choice of divergence is fundamentally a *regularization* decision: the generator function f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$, with Forward KL up-weighting regions where the student underestimates the teacher (coverage) and Reverse KL up-weighting regions of overestimation (precision). Wu et al. (2025) refine this picture for discrete LLM distributions, demonstrating that under practical epoch budgets the mode-seeking/covering dichotomy is better understood through *head-tail focus*: Forward KL concentrates learning on the head of the teacher distribution while Reverse KL emphasizes the tail, explaining why adaptive methods that switch between divergences based on local geometry consistently outperform any fixed choice. At a more foundational level, Cha & Cho (2025) identify the minimal conditions under which a student can provably benefit from teacher supervision versus learning from data alone: the benefit scales with the *information gap* between the teacher's distribution and the data distribution as seen by the student, providing a theoretical foundation for the "exploitable gap" identified empirically by Li et al. (2026d). A complementary unification comes from Li et al. (2026b), who prove that Group-Relative Policy Optimization (GRPO) and self-distillation are mathematically equivalent under certain conditions, with the preference signal in GRPO being implicitly a form of self-distillation against the model's own best rollouts. This suggests that the choice between GRPO-style RL and explicit self-distillation is often a matter of implementation convenience rather than fundamental algorithmic difference.

A persistent practical concern is *length inflation*: OPD students' outputs grow progressively longer during on-policy training. Luo et al. (2026) identify the root cause as an asymmetry in the KL gradient, where positive KL gradients (pushing toward teacher-preferred tokens) are stronger than negative ones (pulling away from dispreferred tokens), creating a net bias toward longer sequences. Their Stable-OPD framework introduces a reference divergence term anchoring the student's length distribution to a pre-distillation baseline, combined with rollout mixing that blends on-policy student generations with off-policy teacher generations in a decaying ratio. These stabilization mechanisms achieve +7.2% over vanilla OPD by breaking the self-amplification cycle, and are composable with any divergence choice.

Taken together, the methods in this section reveal a progression from *local to global* signal management. Token weighting (TIP, SelecTKD, AdaKD) addresses within-sequence heterogeneity. Rollout routing (SCOPE) addresses across-trajectory heterogeneity. Stability mechanisms (Stable-OPD) address temporal dynamics across the entire training run. Curriculum design (PACED) addresses prompt-level selection. Each layer addresses a distinct source of inefficiency, and they compose naturally: one can select tokens by Soft-OR score *within* rollouts that have been routed by quality *within* a training loop that is stabilized against length inflation *across* prompts sampled by curriculum difficulty. This composability is not accidental: the four layers operate at orthogonal granularities (token, sequence, step, prompt), so combining them produces multiplicative rather than redundant efficiency gains.

G-OPD's equivalence proof (Section 4.3) establishes that standard OPD is a special case of dense KL-constrained RL, bridging the distillation and RL communities and ensuring that advances in either field immediately transfer to the other. This equivalence extends naturally to Direct Preference Optimization (DPO): under DPO's closed-form solution, the optimal policy satisfies $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \exp(r(y,x)/\beta)$. Substituting the teacher's log-probability as the implicit reward ($r(y,x) = \log P_T(y|x)$) yields $\pi^*(y|x) \propto \pi_{\text{ref}}(y|x) \cdot P_T(y|x)^{1/\beta}$, which is precisely the geometric mixture distribution targeted by interpolated methods like GKD with λ-mixing. This reveals that DPO-based distillation methods (AlignDistil, ORPO-Distill) implicitly optimize toward the same geometric mixture as token-level KL methods, differing only in their optimization trajectory (contrastive pairs vs. direct gradient descent), and are most useful when constructing preference pairs is cheaper than computing full teacher logits.

The unified KD+RL framework of Li et al. (2025) makes this complementarity precise by decomposing the hybrid gradient into a dense KD component (stable, analytically computable) and a Monte Carlo RL component (high variance, reward-driven): KD dampens variance in policy gradient estimation while RL prevents collapse onto suboptimal teacher modes, with the optimal KD-to-RL weighting ratio decreasing over training as the student's policy stabilizes. Song et al. (2026) further show that textual feedback from a teacher can expand the effective capabilities of RL beyond what outcome-only rewards achieve, providing theoretical justification for the verbal feedback methods (Lion, OVD, ThinkTuning) surveyed in Section 5.2: text feedback provides a denser reward signal than binary outcomes, enabling the student to learn from partial successes that binary verification would discard.

Together, these perspectives converge on a central insight: on-policy distillation, reinforcement learning, and preference optimization are not separate training approaches but rather points on a continuous spectrum parameterized by the density of supervision, the choice of divergence, and the source of the training signal.

7.4 On-Policy vs. Off-Policy: The DeepSeek-R1 Analysis

The theoretical frameworks above establish that on-policy training offers principled advantages. But theory alone cannot settle the question of when practitioners should invest in on-policy infrastructure versus relying on simpler off-policy pipelines.

The DeepSeek-R1 anomaly. DeepSeek-R1 (DeepSeek-AI et al., 2025) represents the most prominent off-policy distillation success, training students from 1.5B to 70B on $\sim$800K samples of long chain-of-thought reasoning traces via purely off-policy SFT. On AIME 2024 (pass@1), R1-Distill-Qwen-7B reaches 55.5%, and R1-Distill-Qwen-32B reaches 72.6%. Crucially, off-policy distillation *decisively outperforms direct RL* for small-to-medium models: at the 32B scale, GRPO applied directly to Qwen2.5-32B-Base achieves only 47.0% on AIME 2024, a gap of 25.6 percentage points. This suggests that for small models, the *knowledge transfer* component of distillation (learning reasoning patterns) dominates the *distributional alignment* component (matching the student's own generation characteristics), explaining why off-policy suffices for small students but on-policy becomes necessary as student capacity grows and its generation distribution increasingly diverges from the training data.

Multi-teacher theoretical foundation. The industrial pattern of distilling from multiple specialized teachers (Qwen3, KAT-Coder-V2, Nemotron-Cascade 2) has an emerging theoretical justification. When a student distills from K specialized teachers $\{T_1, \ldots, T_K\}$ each expert in domain $\mathcal{D}_k$, the resulting student approximates a *mixture-of-experts distribution* without requiring MoE architecture. Formally, the multi-teacher objective $\sum_{k=1}^{K} w_k D_{\mathrm{KL}}(P_{T_k} \parallel P_\theta)$ produces a student that interpolates between teacher distributions, with the weights w_k controlling the relative emphasis. On-policy generation is critical here because the student must learn to *route* between reasoning strategies depending on input characteristics, a capability that static off-policy data cannot teach since the routing decision depends on the student's internal state. This explains why ORBIT's multi-budget fusion and KAT-Coder-V2's domain consolidation both require on-policy training: the *conditional strategy selection* is itself a learned capability that emerges only through on-policy exploration.

Three factors explain why off-policy works so well here: (1) the teacher is an exceptionally strong 671B MoE model whose reasoning traces cover a wide region of the problem space, providing such thorough coverage that exposure bias is empirically mitigated for most standard reasoning paths. (2) The $\sim$800K traces include diverse reasoning strategies (backtracking, verification, alternative approaches), effectively exploring multiple paths per problem. (3) The student models are relatively small (1.5B–70B), so the effective complexity of the conditional distribution (given long reasoning prefixes) is manageable for even off-policy learning.

Distillation vs. direct RL: a formal comparison. The decision between knowledge distillation and direct RL (e.g., GRPO, PPO with outcome reward) is not binary but depends on three factors: (1) *teacher quality relative to the reward signal*, where distillation is preferred when the teacher provides richer per-token information than sparse outcome rewards can encode, (2) *student scale*, where smaller students ($\leq$7B) benefit more from dense teacher supervision because their limited capacity cannot efficiently explore reward landscapes, while larger students ($\geq$32B) have sufficient capacity for RL exploration, and (3) *target ceiling*, where distillation is bounded by the teacher's capability while RL can theoretically exceed any fixed teacher through exploration. The hybrid approaches surveyed in Section 4.3 (G-OPD, KDRL, RLAD) resolve this tension by combining both: using teacher logits as a dense baseline reward that reduces variance while allowing sparse outcome rewards to drive exploration beyond the teacher's ceiling. Empirically, this combination consistently outperforms either approach alone across all student scales.

Why on-policy remains necessary. DDT (Zhang et al., 2026c) provides formal justification: the key factor driving RL's generalization advantage is on-policy data generation, which keeps training aligned with the model's evolving distribution. The off-policy ceiling, the maximum achievable performance regardless of data volume, is strictly lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This ceiling gap widens with three factors: task complexity (more reasoning steps ⇒ more paths to cover), output diversity requirements (creative tasks vs. math), and model capability (stronger students explore further from the training data).

The hybrid pipeline. Qwen3 (Yang et al., 2025a) and MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) operationalize this insight, combining off-policy cold-start with on-policy refinement. This two-phase pattern, off-policy SFT to establish a strong foundation followed by on-policy OPD or RL to close the remaining distribution gap, has emerged as the standard industrial recipe. The optimal transition point (when to switch from off-policy to on-policy) depends on the student's initial capability: weaker students benefit more from extended off-policy warmup, while stronger students can transition earlier.

Compute-quality tradeoff. The expected compute cost for N tokens:

$$C_{\text{off}} \approx N \times (F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{24}$$

$$C_{\text{on}} \approx N \times (G_{\text{student}} + \lambda F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{25}$$

where F, B denote forward and backward FLOPs, G is the autoregressive generation cost, and $\lambda \in (0, 1]$ is the teacher supervision refresh rate. Because $G_{\text{student}} \gg F_{\text{student}}$, on-policy training is typically 4–5× more expensive. For generic knowledge transfer, this overhead is rarely justified. For reasoning, code generation, and complex mathematics, where the off-policy ceiling is low and the on-policy marginal return is high, the investment pays off decisively.

Let $\mathcal{U}(\theta, C)$ represent the student's downstream utility given compute budget C. The constrained optimization is:

$$\max \mathcal{U}(\theta, C) \quad \text{subject to } C = \gamma C_{\text{on}} + (1-\gamma) C_{\text{off}} \leq C_{\max} \tag{26}$$

where γ is the fraction processed on-policy. For generic knowledge transfer, $\gamma \to 0$ suffices. For reasoning, $\frac{\partial \mathcal{U}}{\partial C_{\text{on}}} \gg \frac{\partial \mathcal{U}}{\partial C_{\text{off}}}$, justifying the compute multiplier. A natural hybrid curriculum begins with off-policy warmup ($\gamma = 0$) to establish latent representations, then progressively shifts toward $\gamma \to 1$ in later stages as the student's distribution diverges from the static data.

8 Applications, Systems, and Emerging Domains

The preceding sections developed OPD from theory through methods to unified understanding. This section examines how these algorithmic advances translate into practice. The transition from academic prototypes to industrial deployment reveals that the choice of OPD method is ultimately driven not by theoretical elegance but by practical constraints: available compute, teacher access level, the need for multi-teacher coordination, system-level throughput bottlenecks, and domain-specific challenges (cross-modal transfer, privacy). We organize industrial applications by their *deployment pattern* and emerging domains by the new challenges they introduce.

8.1 Industrial Deployment

Recent foundation models have increasingly incorporated OPD into their post-training pipelines, revealing three distinct industrial use patterns.

Two-phase distillation pipelines. The most common industrial pattern combines off-policy cold-start with on-policy refinement. Qwen3 (Yang et al., 2025a) employs strong-to-weak distillation where larger models (32B, 235B-A22B) provide logit-level supervision on student-generated on-policy sequences, with the student generating responses in either thinking or non-thinking mode. The key architectural choice is the "mode fusion" objective: the

Table 2: Comprehensive comparison of on-policy distillation methods organized by primary contribution dimension.

Method	Year	Category	Divergence/Objective	Signal	Granularity	Key Innovation
Objective: Fixed Divergences						
GKD (Agarwal et al., 2024)	2024	Objective	F-KL / R-KL / JSD	White-box	Token	Unified OPD framework, DAgger for LLMs
MiniLLM (Gu et al., 2024)	2024	Objective	Reverse KL	White-box	Sequence	REINFORCE with teacher as reward
DistiLLM (Ko et al., 2024)	2024	Objective	Skew KL (SKL+SRKL)	White-box	Token	α-smoothing for KL stability
DistiLLM-2 (Ko et al., 2025)	2025	Objective	Contrastive SKL/SRKL	White-box	Token	Asymmetric loss per data source
KETCHUP (Fan et al., 2025)	2025	Objective	Reverse KL	White-box	Sequence	K-step truncated REINFORCE
Constrained (Zimmer et al., 2025)	2025	Objective	Constrained KL	White-box	Sequence	State-augmented CMDP
Objective: Adaptive Divergences						
AKL (Wu et al., 2025)	2025	Objective	Adaptive FKL+RKL	White-box	Token	Ratio-based confidence routing
ToDi (Jung et al., 2025)	2025	Objective	Entropy-routed FKL/RKL	White-box	Token	Hard entropy threshold switching
Entropy-Aware (Jin et al., 2026)	2026	Objective	Continuous FKL+RKL blend	White-box	Token	Smooth entropy interpolation
Objective: RL-Augmented						
G-OPD (Yang et al., 2026c)	2026	Objective	KL-Constrained RL	White-box	Token/Seq	OPD $\equiv$ dense KL-RL, reward extrapolation
RLKD (Xu et al., 2025b)	2025	Objective	KD + Reward Model	White-box	Sequence	Structure-aware reward model
KDRL (Xu et al., 2025a)	2026	Objective	Joint KD + RL	White-box	Token/Seq	On-policy KL regularizer during RL
RLAD (Zhang et al., 2026f)	2026	Objective	Trust Region Ratio	White-box	Token/Seq	PPO-style selective teacher following
AlignDistil (Zhang et al., 2025a)	2025	Objective	DPO + Reverse DPO	White-box	Sequence	Synthetic preference from rollouts
SuperCorrect (Yang et al., 2025b)	2025	Objective	Template + Cross-DPO	White-box	Sequence	Hierarchical thought scaffolding
SCoRe (Lyu et al., 2025)	2025	Objective	Earliest-error correction	White-box	Sequence	Root-cause error targeting
Signal: White-Box Innovations						
DSKD (Zhang et al., 2025b)	2025	Signal	Dual-Space KL	White-box	Token	Cross-vocabulary projection
Cross-Tok. KD (Boizard et al., 2025)	2025	Signal	Latent OT alignment	White-box	Token	Optimal transport for tokenizer mismatch
Delta-KD (Cao et al., 2025)	2025	Signal	Base-to-Instruct delta	White-box	Token	Signal isolation
PromptKD (Kim et al., 2024)	2024	Signal	Prompt-based elicitation	White-box	Token	Input-side distillation
TAID (Shing et al., 2025)	2025	Dynamics	Temporal target interpolation	White-box	Token	Progressive target revelation
MiniPLM (Gu et al., 2025)	2025	Signal	Pre-training KD	White-box	Token	OPD during pre-training
Training Dynamics						
TIP (Xu et al., 2026b)	2026	Dynamics	Teacher reliability filter	White-box	Token	Prefix discrepancy attenuation
SCOPE (Zheng et al., 2026)	2026	Dynamics	Rollout routing	White-box	Token	Redirect on flawed prefix
SelecTKD (Huang et al., 2025)	2025	Dynamics	Confidence filtering	White-box	Token	Top-1 probability threshold
AdaSwitch (Peng et al., 2025)	2025	Dynamics	Exploration/guidance switch	White-box	Token	Binary mode switching
PACED (Xu et al., 2026a)	2026	Dynamics	Beta-kernel curriculum	White-box	Sample	Frontier difficulty sampling
Fast OPD (Zhang et al., 2026a)	2026	Dynamics	Prefix truncation	White-box	Hybrid	Early-token focus
Lightning OPD (Wu et al., 2026)	2026	Dynamics	Offline caching	White-box	Token	4$\times$ cost reduction
Speculative KD (Xu et al., 2025c)	2025	Dynamics	Draft-verify	White-box	Block	Amortized teacher cost
Revisiting OPD (Fu et al., 2026)	2026	Dynamics	Top-p rollout + masking	White-box	Token	Empirical failure mode fixes
Stable-OPD (Luo et al., 2026)	2026	Dynamics	R-KL + reference div.	White-box	Token	Rollout mixing for stability
Veto (Jang et al., 2026)	2026	Dynamics	Geometric bridge KL	White-box	Token	Adaptive gradient veto
Self-Distillation						
SPIN (Chen et al., 2024)	2024	Self-Play	Self-play DPO	Self	Sequence	Iterative self-vs-reference game
IRIS (Anonymous, 2026b)	2026	Self-Play	Interpolative Rényi self-play	Self	Sequence	Unifies SPIN/SPACE/SPIF
OPSD (Zhao et al., 2026b)	2026	Self (PI)	KL on GT-conditioned	Self	Token	Ground-truth as privileged info
CRISP (Sang et al., 2026)	2026	Self (PI)	Conciseness PI	Self	Token	57% token reduction, +9% accuracy
GATES (Stein et al., 2026)	2026	Self (PI)	Gated consensus KL	Self	Token	Document as PI, gated supervision
OPSDL (Anonymous, 2026a)	2026	Self (PI)	Short-ctx $\rightarrow$ long-ctx	Self	Token	Long-context self-distillation
SD-ZERO (He et al., 2026)	2026	Self (EF)	Self-revision preference	Self + Verifier	Sequence	Binary reward $\rightarrow$ dense self-supervision
π-Play (Zhang et al., 2026e)	2026	Self (PI+SP)	Multi-agent co-evolution	Self	Token	Internally generated PI from self-play
SSD (Zhang et al., 2026d)	2026	Self (SP)	Temperature-sampled SFT	Self	Sequence	Minimalist self-play
SDPO (Hübotter et al., 2026)	2026	Self (EF)	Textual feedback + DPO	Self + Verifier	Sequence	Credit assignment from outcomes
RLTF (Song et al., 2026)	2026	Self (EF)	Critique-conditioned	Self + Verifier	Sequence	Text feedback for self-play
SRPO (Li et al., 2026b)	2026	Self (EF)	GRPO + SDPO routing	Self + Verifier	Sequence	Sample-routed multi-objective
PRISM (Wang et al., 2026b)	2026	Self (EF)	Black-box pre-alignment	Black-box (API)	Sequence	Multimodal OPD before RLVR
PAINT (Tan & Hong, 2026)	2026	Self (PI)	Partial-solution masking	Self (re-score)	Token	Overlap-adaptive self-distillation
Black-Box						
GAD (Ye et al., 2025)	2025	Signal	Adversarial matching	Black-box	Sequence	Discriminator as proxy reward
Lion (Jiang et al., 2023)	2023	Signal	Verbal feedback	Black-box	Sequence	NL critique curriculum
OVD (Xiong et al., 2026)	2026	Signal	Verbal scores (0–9)	Black-box	Sequence	Score-based trajectory matching
DAIL (Mendes et al., 2026)	2026	Signal	Inverse RL	Black-box	Sequence	Inferred reward from text
LUFFY (Yan et al., 2025)	2025	Signal	Mixed-policy GRPO	Off-policy + RL	Sequence	Importance-weighted off-policy
ORPO-Distill (Singh et al., 2025)	2025	Signal	Preference optimization	Black-box	Sequence	Teacher-ranked pairs

student must learn to produce both short, direct answers (non-thinking mode) and extended chain-of-thought reasoning (thinking mode) by distilling from teachers operating in each mode independently. On-policy generation is essential here because the student's mode-selection behavior (when to think vs. when to answer directly) is itself a learned policy that cannot be captured by static datasets. Qwen3 reports that the on-policy refinement phase accounts for approximately 80% of total training compute but provides the majority of the quality improvement on reasoning benchmarks, with AIME 2024 scores improving by 15–25 absolute points over the off-policy baseline depending on student scale. Gemma 2 (Gemma Team et al., 2024) embeds knowledge distillation directly into pre-training to bridge performance gaps between parameter tiers (27B$\rightarrow$9B$\rightarrow$2B), representing a more aggressive integration where OPD is not a post-training add-on but a core training ingredient from the start. MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) combines distillation from multiple domain-specialized teachers with RL-based training, achieving frontier performance on mathematical reasoning with a 309B MoE model (15B active parameters).

OPD as model consolidation. A second pattern uses on-policy distillation for *merging* multiple specialized experts into a single deployable model. KAT-Coder-V2 (Li et al., 2026a) decomposes agentic coding into five expert domains, each undergoing independent SFT and RL training, and consolidates them via on-policy distillation, achieving 79.6% on SWE-bench

Table 3: Representative experimental configurations of OPD methods across different application domains.

Method	Teacher	Student	Training Signal	Key Benchmarks
Mathematical Reasoning				
G-OPD (Yang et al., 2026c)	Qwen2.5-32B-Instruct	Qwen2.5-Math-1.5B/7B	Dense logit + reward extrap.	MATH, GSM8K, AIME
OPSD (Zhao et al., 2026b)	Self (GT-conditioned)	Qwen2.5-Math-1.5B/7B	GT-PI self-distillation	AIME, HMMT
PACED (Xu et al., 2026a)	Qwen2.5-32B	Qwen2.5-Math-1.5B	Beta-kernel curriculum	MATH, GSM8K
KDRL (Xu et al., 2025a)	Skywork-OR1-Math-7B	R1-Distill-Qwen-1.5B	Joint KD + RL	AIME, MATH
LUFFY (Yan et al., 2025)	DeepSeek-R1 (off-policy traces)	Qwen2.5-Math-7B/1.5B	Mixed-policy GRPO	AIME, AMC, MATH-500
SCoRe (Lyu et al., 2025)	Qwen2.5-72B-Inst	Qwen2.5-7B/3B, Llama-3.1-8B	Earliest-error correction	AIME, MATH, Multi-hop QA
Instruction Following & General				
GKD (Agarwal et al., 2024)	T5-XXL (11B)	T5-Small/Base	On-policy FKL/RKL/JSD	XSum, WMT
DistiLLM (Ko et al., 2024)	GPT-2 XL (1.5B)	GPT-2 (124M)	Skewed KL	ROUGE-L, BERTScore
AlignDistil (Zhang et al., 2025a)	Qwen2.5-72B-Inst	Qwen2.5-1.5B-Inst	DPO + token-level KD	AlpacaEval, MT-Bench
Industrial Scale				
Qwen3 (Yang et al., 2025a)	Qwen3-32B/235B-A22B	Qwen3-0.6B–14B	On-policy logit distillation	AIME, MATH, LiveCode
Gemma 2 (Gemma Team et al., 2024)	Gemma 2 27B	Gemma 2 9B / 2B	KD in pre-training	MMLU, HellaSwag
MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)	Domain specialists	MiMo-V2-Flash (309B/15B MoE)	Multi-teacher logit + reward	AIME, MATH, GPQA
KAT-Coder-V2 (Li et al., 2026a)	5 domain specialists	Unified agentic coder	Specialize-then-Unify OPD	SWE-bench (79.6%)
Nemotron-Cascade 2 (Yang et al., 2026d)	Domain RL/SFT teachers	30B MoE (3B active)	Cascade RL + domain OPD	IMO, IOI, ICPC
ORBIT (Liang et al., 2026)	Multi-mode RL experts	DeepSeek-R1-1.5B, Qwen3-4B	Multi-teacher mode fusion	AIME, GPQA, MMLU-Pro
Multimodal & Domain-Specific				
VOLD (Bousselham et al., 2025)	Qwen3-8B (text-only)	Qwen2.5-VL-3B	GRPO + token-level KL	MMMU-Pro, MathVision
CORD (Hu et al., 2026)	Text-mode LALM	Audio-mode LALM	Cross-modal R-KL + GRPO	Audio QA/reasoning
Speculative KD (Xu et al., 2025c)	Gemma-7B-IT, Qwen2-7B	Gemma-2B, Qwen2-0.5B	Block-verified on-policy KL	Translation, Dialogue
Cross-Tok. KD (Boizard et al., 2025)	Llama-2-7B, Mistral-7B	OPT-350M, Pythia-160M–1B	Latent OT alignment	QA, Summarization
DP-OPD (Khadem et al., 2026)	GPT-2 Large 774M	DistilGPT-2 82M	GKD + DP-SGD (student only)	Yelp, BigPatent
Veto (Jang et al., 2026)	Qwen2-7B-IT	Qwen2-0.5B-IT	Geometric bridge KL	Reasoning, Generation
TAID (Shing et al., 2025)	GPT-2 XL, Llama-3-8B	GPT-2 Base/Med, Llama-3-2B	Temporal interpolation	Summarization, QA
GUI-SD (Chen et al., 2026)	Self (visual PI)	Qwen3-VL-Inst-8B	Entropy-guided KL	ScreenSpot, RefExp, VisGround
Self-Distillation				
SPIN (Chen et al., 2024)	Self (iter $t-1$)	Zephyr-7B-SFT	Self-play DPO	Open LLM, MT-Bench
CRISP (Sang et al., 2026)	Self (concise prompt)	Qwen3-8B/14B	Per-token R-KL	AIME, MATH-500
SD-ZERO (He et al., 2026)	Self (revision-conditioned)	Qwen3-4B-Inst, OLMo-3-7B	On-policy self-distill	AIME, MATH, Codeforces
π-Play (Zhang et al., 2026e)	Self (QCP-conditioned)	Qwen3-4B/8B	Multi-agent self-distill	NQ, TriviaQA, HotpotQA
SSD (Zhang et al., 2026d)	Self (temperature-sampled)	Qwen3-30B-Inst	On-policy SFT	LiveCodeBench v6
SDPO (Hübotter et al., 2026)	Self (textual feedback)	Qwen3-8B	Credit-assignment self-distill	Code, Math
SRPO (Li et al., 2026b)	Self (SDPO+GRPO)	Qwen3-8B	Sample-routed dual-objective	AIME, MATH, Code
Black-Box				
GAD (Ye et al., 2025)	GPT-5-Chat (API only)	Qwen2.5-14B-Inst	Adversarial minimax	LMSYS-Chat, Dolly
Lion (Jiang et al., 2023)	ChatGPT (API only)	LLaMA-7B/13B	NL critique curriculum	BIG-Bench Hard, AGIEval
OVD (Xiong et al., 2026)	QwQ-32B / Env. Agent	Qwen2.5-3B, Llama-3.2-3B	Verbal scores (0–9)	Web QA, MATH

Verified. Nemotron-Cascade 2 (Yang et al., 2026d) applies the same principle to a 30B MoE model with only 3B activated parameters, achieving Gold Medal-level performance on IMO, IOI, and ICPC World Finals with 20× fewer parameters than DeepSeek-V3.2-Speciale.

OPD for multi-budget reasoning. ORBIT (Liang et al., 2026) addresses the deployment challenge of variable reasoning depth: users need different compute budgets depending on query difficulty. It produces stage-wise expert policies through RLVR under progressively tighter context budgets ($L_{k+1} = L_k/2$), then fuses them into a single model through mode-aware Reverse KL:

$$\mathcal{L}_{\text{fusion}}(\phi) = \mathbb{E}_{k\sim U(1,K)} \mathbb{E}_{\substack{q\sim\mathcal{D} \\ o\sim\pi_\phi(\cdot|q,p_k)}} \left[\sum_t \log \frac{\pi_\phi(o_t|q, p_k, o_{<t})}{\pi_{\theta_k}(o_t|q, p_k, o_{<t})} \right] \tag{27}$$

where p_k is the mode-specific prompt and $\{\pi_{\theta_k}\}$ are the frontier experts. Model merging initializes the student to mitigate the cold-start problem identified by Jang et al. (2026): without initialization near the expert manifold, the student's on-policy rollouts rarely reach the teachers' high-density regions, starving the distillation objective of useful signal. Across DeepSeek-Distill-Qwen-1.5B, Qwen3-4B, and Openmath-Nemotron-7B, ORBIT produces models that smoothly interpolate between four reasoning modes (2K to 32K token budgets) with competitive performance at each budget level, matching the initial model's accuracy at the highest budget while providing 2–7× token savings at lower budgets. The connection to the consolidation pattern above is instructive: where KAT-Coder-V2 and Nemotron-Cascade 2 merge experts specialized by *domain*, ORBIT merges experts specialized by *compute budget*, suggesting that on-policy distillation is emerging as a general-purpose multi-teacher fusion tool for any axis of model specialization.

Agentic distillation. OpenClaw-RL (Wang et al., 2026c) extends OPD to the agentic setting where the student must learn long-horizon tool use and environmental state tracking. Using hindsight-guided OPD with a Process Reward Model (PRM), it trains a Qwen3-4B agent that operates across personal, terminal, GUI, and SWE environments. Skill-SD (Wang et al., 2026a) applies importance-weighted Reverse KL with GRPO for skill-conditioned agentic learning, achieving +14.0% on AppWorld and +10.9% on Sokoban when the student encounters new task variants not seen during distillation.

These four patterns reflect a broader shift in how industry views OPD. The first pattern treats it as a *quality refinement* tool (making an existing pipeline produce better outputs). The second treats it as an *architectural simplification* tool (collapsing a complex multi-expert system into a single deployable model). The third treats it as an *inference-time control* mechanism (enabling a single model to serve multiple latency-accuracy operating points). The fourth extends it to *capability domains* where traditional training fails (agentic, multi-turn, multi-modal). The convergence of all four patterns toward on-policy generation, rather than static dataset distillation, confirms that the theoretical advantages of distributional alignment translate directly into industrial value.

8.2 Emerging Domains

Multimodal OPD. Vision-language models face unique exposure bias challenges because multimodal interleaved generation must maintain coherence across modalities, and high-quality visual reasoning data is scarce compared to text. VOLD (Bousselham et al., 2025) demonstrates that a *text-only* teacher can effectively guide a VLM student through on-policy distillation: the teacher provides reasoning supervision on student-generated visual reasoning traces without ever seeing the image, leveraging the text-based reasoning structure as a modality-agnostic scaffold. The critical finding is that an SFT cold-start phase is essential for cross-modal distributional alignment: without it, the student's on-policy rollouts in the visual reasoning space are too far from the teacher's text-based distribution for distillation to converge. On MMMU-Pro, VOLD improves Qwen2.5-VL-3B from 24.8% to 32.1%, demonstrating that cross-modal knowledge transfer through OPD can be surprisingly effective even when the teacher has no access to the visual modality.

Video-OPD (Li et al., 2026c) extends to temporal video grounding, where the student must learn to localize events within video sequences. X-OPD (Cao et al., 2026) crosses the text-speech boundary, distilling a text-mode Qwen3-Omni into its own speech mode by generating on-policy speech responses and aligning them with the model's own text-mode reasoning. CORD (Hu et al., 2026) applies cross-modal self-distillation between text and audio modes of the same model using weighted Reverse KL:

$$\mathcal{L}_{\text{CORD}} = \mathbb{E}_{x\sim\mathcal{D}}\mathbb{E}_{\hat{y}\sim\pi_{\text{audio}}(\cdot|x)}\left[\sum_t w_t \cdot D_{\text{KL}}(\pi_{\text{text}}(\cdot|x,\hat{y}_{<t}) \parallel \pi_{\text{audio}}(\cdot|x,\hat{y}_{<t}))\right] \tag{28}$$

where the audio mode generates on-policy rollouts and the text mode provides token-level targets, combined with GRPO for outcome reward maximization. CORD achieves state-of-the-art speech reasoning performance, demonstrating that different modality modes of the same model can serve as effective mutual teachers through on-policy interaction.

KEPO (Yang et al., 2026b) identifies a distinct failure mode in multimodal settings: sparse RLVR fails due to exploration collapse when initial solve rates are near zero. By conditioning preference optimization on teacher knowledge, KEPO provides the dense signal necessary to bootstrap capable VLMs.

Embodied intelligence. VLA-OPD (Zhong et al., 2026) applies Reverse KL on self-generated robot trajectories, distilling expert demonstrations into a Vision-Language-Action student. The action space is continuous rather than discrete, requiring the student to learn precise motor control from the teacher's dense token-level supervision on self-generated action trajectories, significantly improving sample efficiency over RL and robustness over SFT. HY-Embodied-0.5 (Tencent Robotics X and HY Vision Team, 2026) uses multi-stage OPD to compress a 32B expert VLM into a 2B embodied MoT model for spatial reasoning, outperforming similarly sized models on 16 out of 22 benchmarks.

GUI grounding. GUI-SD (Chen et al., 2026) applies on-policy self-distillation to GUI grounding, where the student generates click trajectories and the teacher (the same model with visual privileged context, including bounding box and Gaussian soft masks) provides token-level KL supervision. The entropy-guided distillation selectively weights tokens where the teacher-student divergence is highest, achieving state-of-the-art on 6 GUI grounding benchmarks with Qwen3-VL-Instruct-8B.

The consistent finding across these multimodal domains is that on-policy generation is *even more critical* in multimodal settings than in text-only ones, because the diversity of valid outputs (multiple correct grasping trajectories, multiple valid temporal localizations, multiple acceptable image descriptions) makes off-policy data coverage impractical.

Privacy-preserving OPD. DP-OPD (Khadem et al., 2026) injects differential privacy noise (DP-SGD) into the student's gradients during on-policy training while keeping the teacher frozen, proving that on-policy querying can extract utility while mathematically guaranteeing protection of the teacher's training data. This addresses a growing concern as distillation from proprietary models becomes standard practice: without privacy guarantees, on-policy querying could memorize and reproduce private training examples from the teacher.

Autonomous driving. OPD-AV (Afsharrad et al., 2026) applies GKD with 5× compression for autonomous driving planning, distilling a Qwen3-8B SFT model into a 1.7B student on the nuScenes dataset.

8.3 System-Level Integration

The compute optimization methods surveyed in Section 6.3 address algorithmic efficiency, but industrial-scale OPD also requires *systems-level* solutions that span the full hardware-software stack.

Production OPD pipelines must orchestrate three concurrent workloads: student rollout generation (autoregressive, memory-bound), teacher scoring (compute-bound forward pass), and student gradient update (compute-bound backward pass). Frameworks like OpenRLHF and veRL decompose these into separate process groups with distinct parallelism strategies: teacher inference uses tensor parallelism across 4–8 GPUs for low latency, student generation uses pipeline parallelism for throughput, and the optimizer uses ZeRO-3 for memory efficiency. The key engineering challenge is *synchronization*: since the student's policy changes after each gradient step, all pending rollouts become stale. Asynchronous architectures that tolerate slight policy staleness can achieve 30–50% higher throughput at the cost of slightly noisier gradients. High-throughput serving engines (vLLM, TensorRT-LLM) can be repurposed for the generation phase, with continuous batching and PagedAttention enabling efficient rollout generation across large prompt pools, though white-box OPD additionally requires exposing the full vocabulary logits rather than just the sampled token. The communication challenge is non-trivial: a 70B teacher with 7B student on 8×H100 must transmit $\sim$16 GB of logit data per batch ($[B, T, |V|] \times 2$ bytes in BF16), feasible within a single node via NVLink (900 GB/s) but requiring careful tensor slicing or top-k sparsification for multi-node setups.

8.4 When to Use On-Policy vs. Off-Policy

The theoretical framework above yields concrete engineering guidelines. Rather than treating on-policy training as a universal improvement, practitioners should evaluate whether their specific setting satisfies the conditions that make on-policy worth its compute overhead.

The choice between off-policy and on-policy distillation depends on a clear set of conditions. Off-policy SFT suffices when the target distribution is well-covered by the teacher's beam-search outputs (instruction following, factual QA) and three conditions hold simultaneously: an exceptionally strong teacher ($>$500B), diverse reasoning traces, and a relatively small student ($\leq$70B). DeepSeek-R1 (DeepSeek-AI et al., 2025) exemplifies this regime. The transition to on-policy becomes necessary when (1) the student exhibits clear exposure bias (performing well on teacher-style but not user-generated prompts), (2) the task involves multi-step reasoning where compounding errors degrade quality (Ross et al., 2011; Gudibande et al., 2023), or (3) the student needs to exceed the teacher via reward-guided exploration (Yang et al., 2026c). The formal criterion is whether the inference-time state visitation $d_{\pi_\theta}(s)$ diverges significantly from training-time $d_{\mathcal{D}}(s)$.

Within on-policy methods, the choice between logit-level and reward-level supervision depends on student scale: small students ($<$7B) benefit from dense logit supervision (GKD, DistiLLM) to bootstrap competence, while larger students ($\geq$7B) with reliable verifiers benefit from reward-guided OPD (KDRL (Xu et al., 2025a), G-OPD (Yang et al., 2026c)) that enables discovery beyond the teacher ceiling. The hybrid approach combining off-policy SFT warm-up with on-policy refinement consistently yields the best results across the surveyed literature (Qwen3 (Yang et al., 2025a), MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)). At a deeper level, OPD and RLHF share identical computational structures: on-policy rollouts, per-trajectory signals, policy gradient updates. The distinction is purely the signal source (teacher distribution vs. reward model preference). Methods like G-OPD and REOPOLD (Ko et al., 2026) formalize this equivalence, with practical consequence: any RLHF infrastructure can be directly repurposed for OPD, explaining why frameworks like OpenRLHF serve both communities.

8.5 The Distillation Tax: Compute Budget Allocation

Based on the compute analysis in Section 6.3 and the staged pipelines described in recent industrial reports (DeepSeek-AI et al., 2025; Yang et al., 2025a), a practical rule of thumb emerges for compute budget allocation. The most effective industrial pipelines follow a three-stage pattern:

1. **Off-policy warm-up** (50–70% of budget): Standard SFT on teacher-generated data provides cheap, high-bandwidth learning that establishes the student's basic capabilities and aligns its distribution close enough to the teacher for effective on-policy training.
2. **On-policy logit distillation** (20–35% of budget): Token-level OPD with adaptive divergences (GKD, DistiLLM, ToDi) closes the distribution gap on the student's own generation trajectories, addressing exposure bias where it matters most.
3. **Reward-guided refinement** (10–20% of budget): RL-augmented objectives (G-OPD, KDRL, RLAD) push the student beyond the teacher's capability ceiling on reasoning tasks, using sparse outcome rewards to discover novel solution paths.

This staged approach front-loads cheap, high-bandwidth learning and reserves expensive on-policy compute for the final quality push. Qwen3 (Yang et al., 2025a) exemplifies this pattern, with its four-stage pipeline (pre-train $\rightarrow$ SFT $\rightarrow$ RL $\rightarrow$ long-context) allocating progressively more compute to on-policy methods as training progresses. The key insight is that on-policy training is most valuable *after* off-policy warm-up has brought the student close to the teacher's distribution, minimizing the wasted rollouts that occur when the student-teacher gap is too large (Section 7.1).

9 Open Problems and Future Directions

Despite rapid progress, several fundamental challenges remain unresolved. We organize open problems by their potential impact on the field.

Distillation scaling laws. While parameter-scaling laws (Chinchilla (Hoffmann et al., 2022), Kaplan) are well-established for pre-training, no analogous framework exists for distillation. Busbridge et al. (2025) initiated the study of distillation-specific scaling, fitting parametric curves that reveal optimal teacher size grows sub-linearly with compute budget. However, the relationship between teacher-student size ratios, online rollout volume, divergence choice, and asymptotic performance remains largely unquantified. A natural conjecture is that the distillation loss follows a joint power-law:

$$L(N_S, N_T, D_{\text{on}}) = E + \frac{A}{N_S^{\alpha}} + \frac{B}{N_T^{\beta}} + \frac{C}{D_{\text{on}}^{\gamma}} + f(N_S, N_T) \tag{29}$$

where E is the irreducible task entropy and $f(N_S, N_T)$ models the capacity-gap interference. DeepSeek-R1 (DeepSeek-AI et al., 2025) provides empirical grounding: on AIME 2024, performance scales as 28.9% $\rightarrow$ 55.5% $\rightarrow$ 69.7% $\rightarrow$ 72.6% for student sizes 1.5B $\rightarrow$ 7B $\rightarrow$ 14B $\rightarrow$ 32B, with the steepest gain between 1.5B and 7B (26.6% absolute) and the 14B$\rightarrow$32B jump yielding only 2.9%. This suggests that α (student scale) exhibits rapid diminishing returns while teacher capacity dominates. Qwen3 (Yang et al., 2025a) further suggests that $f(N_S, N_T)$ is non-trivial: distilling from multiple specialized teachers yields better scaling than from a single monolithic teacher of equivalent total parameters. Establishing such scaling laws would transform OPD from an empirical art to a predictable engineering discipline.

Uncertainty-aware feedback. Teachers currently provide point-estimate probabilities. When the teacher itself is uncertain about a token, its logits carry little useful information, yet the student treats all teacher signals equally. Future OPD frameworks must incorporate the teacher's *epistemic uncertainty*, allowing the student to discount feedback on queries where the teacher is guessing. Process reward models (Lightman et al., 2024) provide a potential path: by scoring each reasoning step independently, they could identify precisely where teacher supervision is reliable versus speculative. The connection to the failure-mode analysis of Section 7.2 is direct: the flawed prefix trap is fundamentally a problem of unrecognized teacher uncertainty, and explicitly modeling this uncertainty would provide a principled solution. Concretely, one could augment the OPD objective with a teacher confidence weight $c_t = 1 - H(p_T(\cdot|x, y_{<t}))/\log|V|$, suppressing the gradient at positions where the teacher's entropy approaches maximum. TIP (Xu et al., 2026b) provides empirical support for this direction, demonstrating that teacher-student divergence-based filtering outperforms entropy-only schemes. The next step is to extend this to *predictive* uncertainty: rather than measuring the teacher's entropy on the student's current prefix, estimate what the teacher's reliability *would be* several tokens ahead, enabling the student to avoid committing to reasoning paths that will lead to regions of teacher incompetence.

Agent-level and continual distillation. Current OPD focuses on single-turn generation. Extending to multi-turn agentic settings requires fundamentally new credit assignment mechanisms: the student must learn long-horizon tool use, environmental state tracking, and error recovery across entire interaction trajectories. OpenClaw-RL (Wang et al., 2026c) and Skill-SD (Wang et al., 2026a) represent early steps, but the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026d) poses a fundamental challenge. Complementarily, production models undergo repeated fine-tuning cycles (safety patches, capability updates), raising the question of how distillation objectives should adapt when the teacher itself is being updated. SDFT (Shenfeld et al., 2026) and OEL (Ye et al., 2026a) provide initial answers, but distilling from a non-stationary teacher without catastrophic forgetting remains open.

Efficiency frontiers. The 3–8$\times$ compute overhead of on-policy training remains a barrier. While Fast OPD, Lightning OPD, and Speculative KD have made progress, the theoretical minimum cost of on-policy distillation (the information-theoretic lower bound on how much on-policy data is needed to achieve a given quality level) is unknown. A promising underexplored direction is *selective teacher inference*: MiniPLM (Gu et al., 2025) demonstrates that selecting training instances based on the log-probability discrepancy between teacher and a reference model ("Difference Sampling") substantially improves data efficiency under off-policy KD. Transplanting this principle to on-policy rollouts, using teacher-reference divergence to prioritize which student-generated sequences deserve dense teacher feedback, could yield significant compute savings by avoiding expensive teacher inference on uninformative rollouts where the student has already converged.

Latent-space and cross-modal distillation. Current OPD operates exclusively in vocabulary space, where cross-architecture alignment requires expensive projections (DSKD, Cross-Tokenizer KD). A more fundamental approach would distill in a *shared latent space* bypassing the vocabulary bottleneck entirely, enabling distillation between architecturally dissimilar models that share no structural overlap. The extension to multimodal settings raises additional open questions: How does exposure bias manifest when multiple modalities must maintain coherence? What are the optimal divergence choices when teacher and student operate in different modality spaces? Can the unified f-divergence framework be extended to handle continuous signal spaces (images, audio) alongside discrete tokens? VOLD, X-OPD, and CORD provide empirical starting points, but the theoretical foundations for cross-modal OPD remain to be established.

Privacy and evaluation methodology. DP-OPD (Khadem et al., 2026) provides the first principled framework for differentially private distillation, with an asymmetric design applying DP-SGD only to the student while keeping the teacher frozen. The on-policy formulation is well-suited to the DP setting because training on self-generated prefixes reduces compounding errors from DP noise. Open questions include scaling DP-OPD to reasoning-capable LLMs and integrating DP with self-distillation. On the evaluation front, Fu et al. (2026) demonstrate that token-level OPD can appear successful on standard benchmarks while failing on distribution-shifted prompts, highlighting a gap between benchmark performance and true generalization. The precision-recall framework of Cha & Cho (2025) shows distillation induces a fundamental trade-off between sample precision and distributional coverage, suggesting that future evaluation must probe whether the student has learned underlying causal mechanisms rather than superficial correlations.

Closing the distillation-RL loop. Distillation and RL are historically treated as separate, linearly staged processes. Static distillation saturates when the student exhausts the teacher's knowledge, while pure RL diverges when reward signals are sparse. Chen et al. (2025) show that RL retains more pre-training capabilities than SFT because on-policy rollouts maintain distributional proximity to the student's prior. KDRL (Xu et al., 2025a) jointly optimizes KD and RL, preventing the catastrophic forgetting that plagues sequential pipelines. REOPOLD (Ko et al., 2026) operationalizes the unification, with its relaxed on-policy framework combining dense KD gradients for early-stage stability with RL-style reward signals for late-stage exploration. Principled methods for alternating between objectives, determining when to switch, and ensuring that RL-discovered capabilities are not overwritten during subsequent distillation phases, remain the ultimate algorithmic frontier.

Agentic and lifelong OPD. Extending OPD from single-turn text generation to multi-turn agentic settings introduces three unsolved problems. *First, environment non-stationarity*: agentic environments change in response to actions, making direct trajectory comparison meaningless unless the teacher provides counterfactual evaluations conditioned on the student's actual state. *Second, tool-use combinatorics*: modern agent frameworks expose dozens of tools with structured argument schemas, suggesting distillation may need to operate at the tool-call level rather than individual tokens, with credit assignment mechanisms attributing trajectory success to specific tool selections. *Third, safety-critical exploration*: a single incorrect file write in coding agents or misguided click in web-browsing agents can trigger irreversible actions, requiring safety constraints that prevent exploring catastrophically dangerous action sequences. At a higher level, current OPD treats distillation as a discrete training phase. The rapid pace of capability advancement means students benefit from periodic re-distillation against improving teachers. OEL (Ye et al., 2026a) takes a first step using interaction outcomes as dynamic PI, but the broader challenge of *lifelong distillation*, re-alignment without catastrophic forgetting, without retraining from scratch, and without accumulating distributional drift, connects to continual learning with the added complication that both teacher and student distributions are non-stationary.

10 Conclusion

This survey has systematically examined on-policy distillation for large language models, synthesizing over 90 methods that span foundational formulations, algorithmic innovations, theoretical analyses, and industrial deployments. We organized this rapidly growing literature around a practitioner's decision chain rather than descriptive categories, tracing three fundamental design axes: *objective function* design (from fixed divergences through adaptive routing to RL-augmented objectives that break the teacher ceiling), *signal source* architecture (from dense white-box logits through black-box API access to self-distillation that eliminates the external teacher entirely), and *training dynamics* (token weighting, curriculum adaptation, and compute optimization).

Key Findings. Three central insights emerge from this synthesis. *First*, shifting from off-policy to on-policy training distributions, from $y \sim p_{\text{data}}$ to $y \sim p_\theta$, is a consistently impactful design choice, yielding meaningful accuracy gains across mathematical reasoning, code generation, and instruction following (Tables 2 and 3). The improvement stems from eliminating exposure bias: the student learns to recover from its own errors rather than memorizing idealized trajectories. Across the methods surveyed, on-policy training provides typical gains of 5–15% absolute on reasoning benchmarks over off-policy baselines, with the gap widening for longer reasoning chains. *Second*, divergence selection is not one-size-fits-all but must be adapted to both task and token position. Reverse KL excels at reasoning tasks where mode-seeking prevents hallucination, while Forward KL preserves the diversity needed for open-ended generation. Adaptive methods such as ToDi (Jung et al., 2025) and Entropy-Aware OPD (Jin et al., 2026) that dynamically select divergences based on local distributional geometry represent the current frontier. *Third*, the boundary between distillation and reinforcement learning is rapidly dissolving. G-OPD (Yang et al., 2026c), RLAD (Zhang et al., 2026f), and REOPOLD (Ko et al., 2026) demonstrate that the optimal training signal combines dense teacher supervision with sparse outcome rewards, with KD providing gradient stability while RL enables exploration beyond the teacher's capability ceiling. The unification is not merely conceptual: it provides a shared toolkit (trust regions, advantage estimation, policy gradient variance reduction) that benefits both communities.

Practical Takeaways. With white-box teacher access, token-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) offer the best quality-compute tradeoff for general instruction following. For reasoning-intensive tasks, sequence-level methods (MiniLLM (Gu et al., 2024)) or hybrid approaches (Fast OPD (Zhang et al., 2026a), PACED (Xu et al., 2026a), Lightning OPD (Wu et al., 2026)) are preferable despite higher variance. Lightning OPD demonstrates that offline distillation under teacher consistency can match online OPD at 4$\times$ lower cost, significantly lowering the barrier for academic research. With only API access, GAD (Ye et al., 2025) and OVD (Xiong et al., 2026) enable effective on-policy learning through adversarial and verbal formulations, respectively. Self-distillation methods (SPIN (Chen et al., 2024), OPSD (Zhao et al., 2026b), SD-ZERO (He et al., 2026)) eliminate the need for a separate teacher entirely, making them attractive under resource constraints. The compute overhead of on-policy generation, typically 4–5$\times$ that of off-policy training, can be substantially reduced via speculative decoding (Xu et al., 2025c), prefix truncation (Zhang et al., 2026a), or offline precomputation (Wu et al., 2026).

Looking Ahead. The most pressing open problems are threefold. First, the absence of distillation scaling laws analogous to Chinchilla (Hoffmann et al., 2022) forces practitioners to rely on costly trial-and-error for compute allocation. Second, the field needs uncertainty-aware objectives that prevent the echo chamber effect when students explore out-of-distribution regions where teacher supervision is unreliable. Third, extending OPD to multi-turn agentic settings, where training involves entire interaction trajectories rather than single completions, remains a fundamental challenge compounded by the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026d). As the field matures, we anticipate convergence toward unified KD+RL frameworks that treat distillation not as a separate training stage but as a continuous regularization mechanism throughout the model's lifecycle, from pre-training through deployment. As LLMs transition from static text generators to interactive reasoning agents, the principles of on-policy alignment will only grow in centrality.

Limitations. This survey focuses exclusively on methods that involve on-policy generation from the student during training. We deliberately exclude general knowledge distillation (e.g., feature-based, attention transfer), model compression techniques orthogonal to the training paradigm (pruning, quantization), and methods that use student-generated data only at inference time (self-consistency, majority voting). Our coverage is current through April 2026. Given the field's rapid pace (approximately 8–10 new OPD papers per month), some concurrent work may be absent. Additionally, our practitioner decision tree reflects current best practices and hardware constraints. As compute costs decrease and training frameworks mature, the cost-benefit calculus will shift toward more aggressive on-policy methods.

Reproducibility challenges. A significant concern across the OPD literature is the difficulty of fair comparison. Methods are evaluated on different base models, different training compute budgets, different benchmark versions (e.g., MATH-500 vs. full MATH), and with different numbers of rollouts per prompt. Few papers control for all these variables simultaneously, making cross-method comparison unreliable from published numbers alone. Our Tables 2 and 3 report original-paper results rather than controlled reproductions, and readers should interpret cross-row comparisons with appropriate caution. We advocate for standardized OPD benchmarking protocols that fix the base model, compute budget, and evaluation suite, analogous to what HELM (Liang et al., 2023) provides for general LLM evaluation.

Broader impact. OPD has immediate implications for AI accessibility and safety.

On the accessibility front, effective distillation enables smaller, more efficient models that can run on consumer hardware, democratizing access to advanced reasoning capabilities that would otherwise require expensive API access to frontier models. The self-distillation methods surveyed here (OPSD, SD-ZERO, SSD) are particularly significant: they enable capability improvement without any external teacher, allowing resource-constrained researchers and developers to iterate rapidly.

On the safety front, OPD introduces both risks and mitigations. The risk is that effective distillation of frontier capabilities lowers barriers to misuse. The mitigation is that the teacher retains a supervisory role during training, providing a natural point for injecting safety constraints, alignment objectives, and capability boundaries. As the field matures, we expect safety-aware OPD, where the teacher not only transfers knowledge but also enforces alignment properties during the interactive training loop, to become a critical research direction.

On the environmental front, OPD's compute overhead (4–5$\times$ over off-policy SFT) raises sustainability concerns at industrial scale. However, the resulting smaller, more capable students reduce downstream inference costs by orders of magnitude, and the net carbon footprint over a model's lifetime is typically lower when amortized across billions of inference calls.

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